How Much Work Can AI Agents Do?

Task Value Inside the AI Time-Horizon Frontier

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Abstract

METR evaluates AI agents by the length of task, in human working time, that they can complete on their own. This paper puts that measure on an economic scale. I map human time estimates from the Anthropic Economic Index to 18,796 O*NET tasks and weight each task by the probability of success that METR's curves imply for its length. Tasks carrying 46 per cent of the US wage bill are expected to succeed under the latest frontier model, up from 14 per cent at the start of 2025. Across jobs, coverage is thinner: 19 per cent of US employment is in occupations where at least half of the work is expected to succeed, and no occupation reaches 90 per cent. This capability measure is grounded in measured success rates of AI agents and updates with each new model, so it can track the frontier as it moves and give macroeconomic models of AI a capability-based value for the share of work AI could automate.

JEL codes: E24, J23, J24, O33. **Keywords:** AI agents, time horizons, task-based models, automation, labour markets.

1 Introduction

METR measures the performance of AI agents by the maximum length of the tasks they can autonomously complete, expressed as the time a human would require for the same work (Kwa et al., 2025). On METR's evaluation suite, this time horizon has doubled roughly every 4.2 months since 2023 (METR, 2026). The chart of these horizons is now followed closely by AI researchers, investors and industry observers, who read it for signs that progress is speeding up or slowing down (New York Times, 2026). Over the same period, the Anthropic Economic Index (AEI) has tracked which work tasks users delegate to Claude and estimated how long those pieces of work would take a person without AI (Handa et al., 2025; Anthropic, 2026). This paper places both measures on a single scale, US work tasks weighted by their share of the wage bill, to measure two quantities: the share of the wage bill in tasks a frontier agent is expected to complete, and the share in tasks that workers actually delegate to an AI system.

Macroeconomic models of AI rely on assumed values for both of these quantities. In growth models, the effects of AI turn on the share of tasks it automates (Trammell and Korinek, 2026). In the scenario framework developed at Anthropic by Korinek et al. (2026), the "extreme" 2030 scenario,

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for example, assumes that AI can affect 50 per cent of all tasks (the affected mass m_t), that it is used on 60 per cent of the instances of those tasks (the diffusion share d_t), and that nine in ten AI-performed tasks are automated rather than augmented (the automation share ψ_t). Across the values chosen for these parameters, the framework produces outcomes that range from negligible disruption to 17.9 per cent cognitive unemployment, 15.4 per cent annual GDP growth and a labour share of income of 45.2 per cent. The anchor used so far for the affected mass comes from usage data: it counts the tasks on which AI is already used substantially, which mixes what AI can do with how far its use has spread. This paper provides an empirical value for the automation side of these parameters that rests on capability rather than use: it uses a *capability curve* to measure the share of work AI could perform outright, and traces how that share has grown over time. Any task an agent can complete alone is one that AI affects, so the measure is also a lower bound on the affected mass measured as what AI can do rather than where it is used. If METR's curves carry over to the rest of the economy, this lower bound already exceeds the affected mass assumed in the substantial 2030 scenario, so the spread of outcomes depends mainly on diffusion, the gain per task and the automation share (Section 5.3).

To measure capability, I use METR's fitted logistic curves, which map an agent's completion probability to the logarithm of a task's human time. Existing measures of occupational exposure to AI rate how closely each task relates to what AI can do, from patent text, crowd-sourced ratings or a scoring rubric, and rank occupations at one point in time (Webb, 2020; Felten et al., 2021; Eloundou et al., 2024). The METR curve is measured on agents attempting tasks, gives a probability of success rather than a rank, and has a trend that can be projected forward. In the language of task-based models (Acemoglu and Restrepo, 2018), it orders tasks by the time they take a person and makes the frontier model's time horizon an automation threshold that moves out at a measured rate. Rather than counting only the tasks below a threshold such as the 50 or 80 per cent horizon, I calculate the probability of success for each task given its length and sum the wage bill weighted by that probability. This expected-success share is a continuous version of the threshold measure.

I anchor the estimates on two models: Claude Opus 4.6, the most capable model inside METR's 16-hour reliability range, and Mythos Preview (early), a pre-release checkpoint that is the latest measured frontier but sits above that range. Under Claude Opus 4.6 the expected-success share is 40 per cent of the US wage bill; under Mythos Preview (early) it is 46 per cent. The share has risen from 14 per cent under o1, the frontier at the start of 2025. Its level is highly sensitive to which tasks count as within the scope of an AI agent, which I set out below. Its growth is more robust to those definitions: under every scope rule I consider, the share has grown by a factor of 3 since then, and under stricter standards of success it has grown faster still. Most of that growth came as horizons passed the tasks of a few hours where most of the wage bill sits. Nearly all work in scope is now shorter than the 50 per cent horizon, and one more doubling of the horizon would add about 5 points, so further gains depend more on how reliably agents complete short tasks than on how long a task they can attempt. Tasks delegated to Claude carry between 25 and 30 per cent of the wage bill in each AEI data window, and 49 per cent if every task that has appeared in any window is counted; 18 of the 46 points of expected success are in tasks with no recorded use. Where AI is used, shorter tasks, on which an agent is more likely to succeed, are more often handed to Claude outright than worked on together.

These shares use METR's definition of success, in which an agent completes a well-specified task that is scored automatically. Stricter standards give lower success. When repository maintainers judge whether they would merge an agent's code, rather than relying on its tests, METR finds

horizons about 7 times shorter (Whitfill et al., 2026). At that standard the expected-success share is 27 per cent, equivalent to the frontier of about 12 months earlier. Other success curves can be drawn, for instance from the AEI's own success labels, but these reflect which work users choose to delegate and how they iterate on it (Appel et al., 2026), and success may follow different curves in different domains (Kwa and Cheng, 2025). METR's curve is the most careful measurement available, and applying it outside software is the central assumption of this paper; Section 3 and Appendix I set out what the other curves show.

The shape of the curve is itself in dispute. On METR's curves for frontier models, the log-odds of success fall by 1.4 to 2.3 for each tenfold increase in task length. Mertens et al. (2026) had workers experienced in each occupation grade responses from 41 language models to 6,648 O*NET tasks and find a much flatter curve (Appendix Figure 15): the log-odds that a response is usable without edits fall by 0.27 per tenfold increase, about 7 percentage points at their average success rate of 60 per cent, and newer models raise success by about the same amount at every length. They call the steep pattern a crashing wave and the flat one a rising tide. The shape decides what a frontier means. On a steep curve, the horizon roughly separates the work an agent completes from the work it fails, and the share of the wage bill inside it grows quickly as the horizon passes the tasks of a few hours. On a flat curve, most of the wage bill sits at partial success at any date, a threshold share depends on the probability chosen, and the share starts higher and grows more slowly. A flatter curve does not mean less work within reach. Their curve sits high: at the release of GPT-5 in August 2025, the end of their sample, most of the wage bill in scope is in tasks that their curve expects to succeed 66 to 73 per cent of the time, so it gives a larger share of the wage bill than METR's curves do. I use METR's curves, and Section 5.4 sets out what theirs would change.

The closest work is the January 2026 Anthropic Economic Index report (Appel et al., 2026), which combines per-task success rates classified from Claude conversations with task time shares into an "effective AI coverage" of occupations, and estimates usage-based time horizons for comparison with METR's; its longer usage horizons reflect success that falls more slowly with task length. Massenkoff and McCrory (2026) build an observed-exposure measure from the same usage data, which Korinek et al. (2026) use to anchor their scenarios. This paper differs in using a capability curve measured on a benchmark rather than success classified from usage, in weighting by the wage bill, and in tracing the measure across frontier models since 2023. It builds on task-based models of technical change (Autor et al., 2003; Acemoglu and Restrepo, 2018) and on measures of occupational exposure to AI (Brynjolfsson et al., 2018; Webb, 2020; Felten et al., 2021; Eloundou et al., 2024), which Acemoglu (2025) turns into a macroeconomic estimate by combining exposure with the share of exposed tasks that is cost-effective to automate (Svanberg et al., 2024). Lashkari et al. (2026) model that cost-effectiveness directly, with automation depending on whether a task has enough scale to amortise the fixed cost of adapting a model to it; this paper measures the prior question of which tasks current agents can complete.

Translating benchmarks to the economy requires two extrapolations: AEI's duration estimates for delegated pieces of work stand in for the length of employer-defined tasks, and success curves estimated on software engineering tasks are applied to all work that involves neither physical activity nor live interaction with people. AEI's own success data and worker evaluations of model responses across O*NET tasks (Mertens et al., 2026) suggest that the second is the weaker link (Sections 3 and 5.4). I bound the estimates with a compound bootstrap and a set of robustness checks, and report how far each definition moves the result.

2 Data

2.1 METR time horizons

Figure 1 reproduces METR's headline capability estimates. Each point represents a model's 50 per cent time horizon. METR flags measurements above 16 hours as unreliable because its current suite has few long tasks. Mythos Preview (early) is the only frontier model above this threshold, and Claude Opus 4.6 is the last model measured below it; I report both throughout.

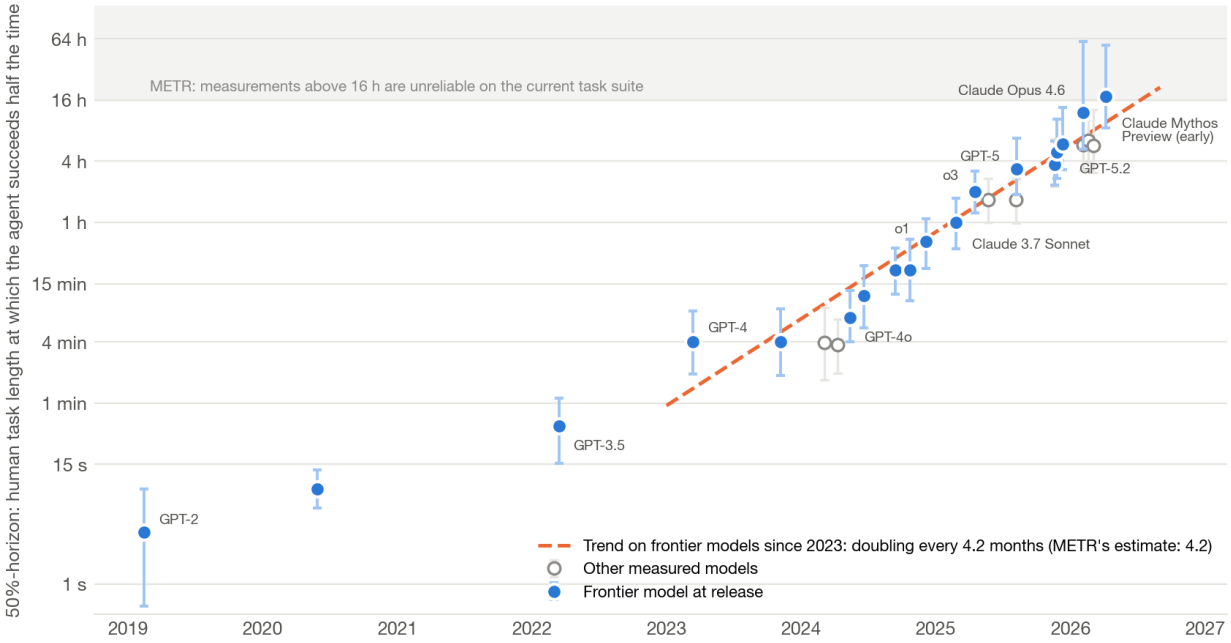


Figure 1: METR's 50 per cent time horizons, reproduced from METR's published estimates (METR, 2026). Bars are METR's published 95 per cent confidence intervals. Filled points: frontier model at release. Dashed line: exponential trend fitted to frontier models released from January 2023 with horizons below 16 hours. Grey band: horizons METR describes as unreliable on the current suite.

METR's evaluation suite is specialised, covering 228 computer-based tasks across 79 families (Appendix Table 7). Most come from HCAST, a set of software engineering, machine learning, cybersecurity and research tasks with completion times benchmarked on human contractors (Rein et al., 2025). To construct the expected-success shares, I rebuilt every run from the raw per-task success counts and refit the logistic curves using METR's methodology. The refit reproduces METR's published horizons to within 0.02 per cent for every model (Appendix Table 8) and supplies the coefficients for the continuous success curve.

2.2 The Anthropic Economic Index

The AEI reports the share of Claude conversations mapped to specific O*NET tasks, alongside an AI-generated estimate of the human time required to complete the delegated work without AI (Handa et al., 2025; Anthropic, 2026). For task durations, I rely on the June 2026 release because it maps tasks by their exact O*NET IDs; earlier releases are retained for robustness checks.

Durations are directly observed for 2,814 tasks, representing 26 per cent of the wage bill. For the

remaining tasks, I impute the duration using the geometric mean of observed durations among tasks sharing an O*NET detailed work activity (see Appendix A for fallback levels and hold-out validation). The resulting wage-weighted median task duration across the economy is 3.1 hours.

The duration of a task depends on how the work reaches Claude. The June release is the first whose Claude.ai data include Cowork, Anthropic's agentic desktop product, and for the same task its durations are 1.6 times those in the February release, while API durations barely change. For the same task in the same month, API durations are 0.26 to 0.50 times Claude.ai durations. The earlier releases give an expected-success share of 49 per cent (Table 13).

2.3 Wage weights from O*NET and OEWS

I map each O*NET task to its parent occupation and assign it a proportional share of the occupation's total wage bill based on OEWS May 2025 employment and mean annual wages (U.S. Bureau of Labor Statistics, 2026; National Center for O*NET Development, 2026). The share is proportional to importance over the occupation's core tasks, using O*NET's own core/supplemental classification. Summed over all 18,796 tasks, this yields $10.72 trillion across 153.4 million jobs. Hours-implied weights, OEWS annualisation conventions and admitting the supplemental tasks all leave the aggregate within a point of the baseline (Table 13).

2.4 Tasks, work activities and the excluded set

A task in this paper is an O*NET task statement, such as "record patients' medical information and vital signs" for registered nurses, and it is the unit that carries the wage weight. Its length is the AEI estimate of the human time for the piece of it that users delegate in a conversation, while success is measured on METR's tasks, which are self-contained and well specified. These units need not coincide; Section 5.5 discusses the consequences.

Which work a software agent could do is a matter of definition. The baseline uses O*NET's own classification of work activities and excludes two kinds of work. A task is excluded if *any* of its detailed work activities maps to a physical generalized work activity (handling objects, controlling machines, operating vehicles, assisting and caring for others, and similar), or to one of the nine generalized work activities that O*NET defines as live interaction with people: establishing relationships, selling or influencing, resolving conflicts and negotiating, working directly with the public, building teams, training and teaching, directing subordinates, coaching, and staffing. O*NET defines the remaining interpersonal activities without reference to a mode — communicating with colleagues, for example, is "by telephone, in written form, e-mail, or in person" — and they stay in scope, as do interpreting information for others, coordinating, advising and administration. A task is in scope only if none of its activities is excluded. Table 1 illustrates this mapping.

The strict rule reflects the premise that the expected-success share measures the completion of a task as defined: a software agent cannot complete a mixed task unless an employer unbundles it and reassigns the physical or interactive part, and METR's tasks are self-contained deliverables, which a meeting, a negotiation or the supervision of staff is not. Of the 18,796 tasks in the economy, 4,819 are entirely physical and 657 mix physical and other activities; a further 2,579 involve no physical activity but include live interaction. Physical tasks carry 22 per cent of the wage bill and the interactive ones 19 per cent; all excluded tasks, 40 per cent of the wage bill, are assigned a success probability of zero.

Table 1: From task to work activity: the O*NET mapping for one task in each class. Data Entry Keyers: a task in scope and an entirely physical task. General and Operations Managers: a task excluded only for live interaction. Construction Laborers: a task that mixes physical and other activities. A task is excluded if any detailed work activity it maps to sits under a physical or live-interaction generalized work activity.

Task (occupation)	Detailed work activities → generalized work activity	Counted as
Compile, sort, and verify the accuracy of data before it is entered. (Data Entry Keyers)	Compile data or documentation. → Processing Information	in scope
Perform personnel functions, such as selection, training, or evaluation. (General and Operations Managers)	Conduct employee training programs. → Training and Teaching Others (live interaction) Hire personnel. → Staffing Organizational Units (live interaction)	excluded
Measure, mark, or record openings or distances to layout areas where construction work will be performed. (Construction Laborers)	Measure work site dimensions. → Estimating the Quantifiable Characteristics of Products, Events, or Information Mark reference points on construction materials. → Handling and Moving Objects (physical)	excluded
Store completed documents in appropriate locations. (Data Entry Keyers)	Store records or related materials. → Handling and Moving Objects (physical)	excluded

Table 2: Task universes. Shares are of the $10.72 trillion wage bill attached to all tasks.

Universe	Tasks	Share of wage bill
All O*NET tasks in occupations with OEWS employment and wages	18,796	100.0%
In-scope tasks (no physical or live-interaction work activity)	10,741	59.5%
Tasks with an observed AEI duration (June 2026 AEI release, Apr and May 2026)	2,814	25.6%
Computer and Mathematical occupations (in-domain check, SOC 15)	770	5.9%

The rule records what a task involves, not where it is done. O*NET files "measure felled logs" under estimating quantities, so log graders and agricultural inspectors pass it. Adding the inspecting work activity to the excluded set moves only 3.2 per cent of the wage bill and leaves both occupations in place. Setting is recorded only at the level of the occupation, so as a bound I also apply the rule of Dingel and Neiman (2020), rebuilt on O*NET 30.2, which excludes every task in occupations that cannot be done remotely. Tasks in scope in such occupations carry 25 per cent of the wage bill. Dingel and Neiman count school teaching as remote (it could be done at home in 2020), so I also exclude occupations that require close physical contact (Appendix J). Figure 2 shows the range: the expected-success share under Mythos Preview (early) runs from 24 per cent with both exclusions (26 per cent with Dingel–Neiman alone) to 52 per cent with a lenient rule that excludes a task only if every activity is physical or interactive. The physical filter alone, which leaves live interaction in scope, gives 60 per cent, and the Eloundou et al. (2024) exposure labels give 49 per cent. The definition of automatable work moves the share by more than a model generation. The aggregate results below use the baseline filter; the results for individual occupations in Section 5.1 use both exclusions.

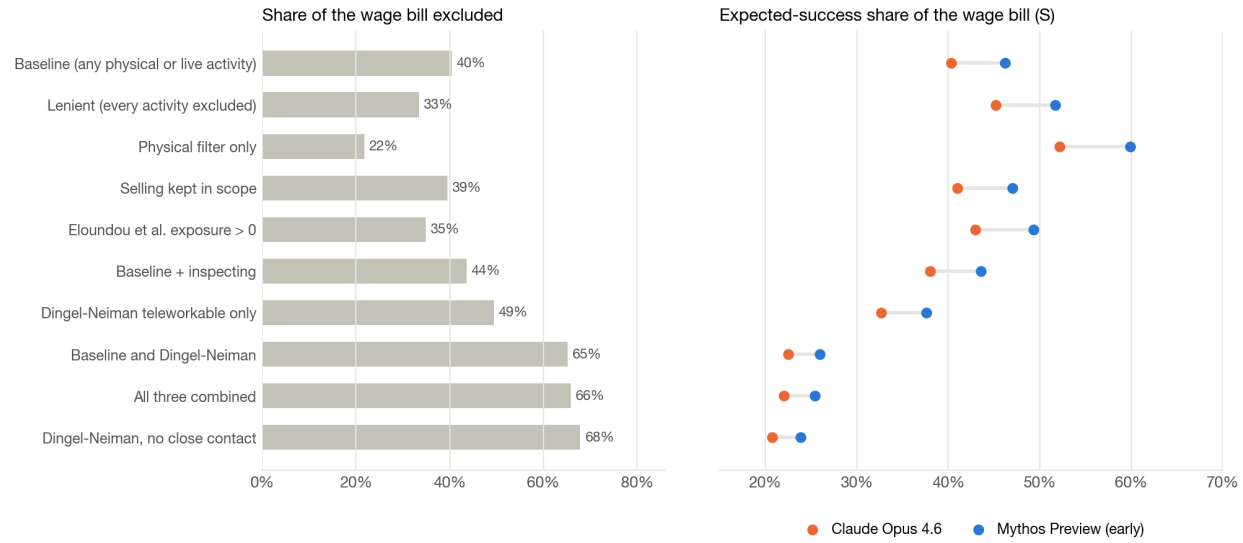


Figure 2: Ten rules for the excluded set. Left: share of the wage bill excluded. Right: expected-success share of the wage bill under both frontier models. Baseline: a task is excluded if any of its work activities is physical or live interaction with people (Section 2.4). Lenient: excluded only if every work activity is. Physical filter only: live interaction kept in scope. Selling kept in scope: generalized work activity 4.A.4.a.6 removed from the interactive set. Eloundou et al.: excluded if the exposure label is zero. Inspecting: tasks with a work activity under generalized work activity 4.A.1.b.2 are also excluded. Dingel–Neiman: every task of an occupation that cannot be done entirely at home is excluded, using their published rule rebuilt on O*NET 30.2 (Appendix J). No close contact: also excludes occupations whose O*NET physical-proximity rating is 4 or more (at arm's length or closer), such as school teaching, which Dingel–Neiman count as remote because it can be delivered through a screen.

3 Measures

Let V_t denote the wage bill attached to task t (Section 2.3), D_t its duration in human hours, and $P_d(D)$ the fitted success curve of the frontier model at date d. The expected-success share is:

$$S_d \;=\; \frac{\sum_t V_t \, P_d(D_t) \, \mathbf{1}[t \text{ in scope}]}{\sum_t V_t}.$$

The threshold shares, F_d^{50} and F_d^{80}, replace the continuous probability $P_d(D_t)$ with an indicator equal to one if D_t is below the 50 or 80 per cent horizon. S_d instead integrates the full curve, giving partial credit to tasks beyond the threshold and discounting shorter tasks by their residual failure rate. It is a measure of capability: it does not account for the cost of checking the agent's work or for the friction of reassigning standing responsibilities to an agent.

Evaluating S_d across the economy requires extrapolating METR's success curves out of domain. The core assumption is that, at a given human duration, an agent's probability of completing a paralegal's task in scope equals the probability predicted for a software engineering task of the same length. Duration is a natural index of difficulty because a longer task chains more steps, each of which can fail (Ord, 2025); the assumption is that the rate of failure per step is the same outside software.

Within METR's suite, task length predicts average success well. For individual frontier models, the weighted R^2 of task success rates on the pooled curve is 0.24 for Mythos Preview (early) and 0.26 for

Claude Opus 4.6. With four to six runs per task, binomial noise caps the attainable R^2 at roughly 0.57, so length accounts for about 43 per cent of the explainable variation (Appendix D). The measure is linear in P_d and therefore requires only that the curve capture the conditional mean of success at a given length, not the outcome of each task. The curve is also stable to the composition of the suite: dropping any single task family moves Mythos Preview (early)'s expected-success share by no more than two points (46 to 47 per cent).

Outside the suite, other success curves can be drawn, but each measures something different from autonomous completion. Two AEI releases report, for each task, the share of conversations that Claude classified as successful alongside their human time. On Claude.ai this share barely changes with duration: across tasks, the fall in the log-odds of success per doubling of human time is between -0.04 and 0.04, against 0.41 to 0.66 for the METR curves of the Claude models that bracket that traffic. On the API it falls, but more slowly, at 0.12 to 0.21 per doubling within occupations and work activities, and at 0.37 within the same task across the two releases (Appendix I). The AEI's authors give reasons to expect these curves to be flatter than METR's (Appel et al., 2026). Users choose which tasks to bring to Claude, so observed success reflects their judgement of what will work, of the cost of setting up the task and of the time it would save; if they avoid tasks they expect to fail, observed success overstates capability. On Claude.ai, each turn of a conversation lets the user correct course, which breaks a long task into shorter steps. Users may also respond to better models by bringing harder versions of the same task. And the labels are Claude's own judgement of whether the user's request was met. GDPval, in which experts grade one-shot deliverables on a fixed set of tasks, avoids the selection of tasks and also shows a flatter gradient, 0.21 per doubling for Claude Opus 4.1 against 0.66 on METR's suite, but its standard is a comparison with an expert's work (Patwardhan et al., 2025). Workers' evaluations of single responses to 6,648 O*NET tasks also give a flatter gradient, 0.08 per doubling pooled over 41 language models and no more than 0.16 for any one of them (Mertens et al., 2026); Section 5.4 discusses them. Success also depends on the kind of work, consistent with a "jagged" frontier (Dell'Acqua et al., 2026): METR's comparison across benchmarks finds that the level of the horizon differs by orders of magnitude between domains, with horizons on visual computer-use tasks 40 to 100 times shorter than on software, although they are growing at similar rates (Kwa and Cheng, 2025). I keep METR's curves, whose gradient is measured on autonomous completion under controlled conditions, and treat the variation of success across tasks as the least secure part of the measure.

Every headline table includes an in-domain check restricted to Computer and Mathematical occupations (SOC 15, 5.9 per cent of the wage bill), where the capability assumption is least strained.

4 Results

Table 3 presents the baseline estimates under both frontier models. Moving from Claude Opus 4.6 to Mythos Preview (early) increases the expected-success share of the US wage bill from 40 to 46 per cent. A compound bootstrap that resamples the METR curve, adds measurement noise to AEI durations and re-runs the imputation yields a 95 per cent interval of 40 to 51 per cent under the latest frontier. The interval reflects sampling uncertainty only.

Between the two models, the share of the wage bill in tasks shorter than the 80 per cent horizon rises sharply, from 8 to 27 per cent. The share in tasks shorter than the 50 per cent horizon, by contrast, is close to its ceiling: nearly every task in scope is shorter than the 50 per cent horizon of either model (Section 4.1).

Table 3: Headline numbers, May 2026. "In scope" = no physical or live-interaction work activity. Durations imputed where unobserved. With w_N the in-scope share of the wage bill and $\bar{P}_N$, $\bar{P}_P$ the wage-weighted mean success probability over tasks in scope and excluded tasks: row 1 = $w_N\bar{P}_N$; row 2 = $\bar{P}_N$; row 3 = $w_N\bar{P}_N + (1 - w_N)\bar{P}_P$. SOC 15 rows provide an in-domain benchmark.

Measure (share of wage bill in…)	Denominator	Frontier model	
		Claude Opus 4.6 (Feb 2026; p50 12.0 h, p80 1.2 h)	Mythos Preview (early) (Apr 2026; p50 17.4 h, p80 3.1 h)
Expected success	all tasks	40%	46%
Expected success	in-scope tasks	68%	78%
Expected success, excluded tasks scored by the curve rather than excluded	all tasks	70%	79%
Shorter than 80% horizon	all tasks	8%	27%
Shorter than 80% horizon	in-scope tasks	14%	45%
Shorter than 50% horizon	all tasks	58%	59%
Shorter than 50% horizon	in-scope tasks	97%	100%
Observed in Claude usage, May 2026	all tasks	30%	30%
Observed in any AEI release, Dec 2024 to May 2026	all tasks	49%	49%
Expected success, in-domain check	SOC 15 tasks	54%	63%
Shorter than 80% horizon, in-domain check	SOC 15 tasks	3%	17%
Observed in any AEI release, in-domain check	SOC 15 tasks	82%	82%

Table 4: Compound 95 per cent intervals from 500 nested bootstrap draws, accounting for METR curve fit uncertainty, AEI duration measurement noise and imputation variance. These are sampling intervals; they do not reflect the choice of success concept or of the excluded set.

Measure	Error source resampled	Claude Opus 4.6 point (95% interval)	Mythos Preview (early) point (95% interval)
Expected success, all tasks	METR curve (family, task, run)	40% (35 to 46)	46% (41 to 51)
	Durations (measurement noise and imputation)	40% (40 to 40)	46% (45 to 46)
	Both	40% (34 to 46)	46% (40 to 51)
Shorter than 80% horizon, all tasks	METR curve (family, task, run)	8% (1 to 21)	27% (11 to 48)
	Durations (measurement noise and imputation)	8% (6 to 10)	27% (23 to 29)
	Both	8% (1 to 22)	27% (12 to 46)
Expected success, in-scope tasks	Both	68% (57 to 77)	78% (67 to 85)

4.1 The task duration distribution

Figure 3 shows why the threshold shares behave so differently. Most of the wage bill falls between 15 minutes and 16 hours, and nearly all value in scope lies below 12.0 hours, the shorter of the two 50 per cent horizons, so the p_{50} threshold captures nearly all tasks in scope.

This near-saturation reflects the AEI duration data, not a ceiling on capability. AEI durations measure modular pieces of work delegated in chat, which are short by selection. Being shorter than the p_{50} horizon only means that the fitted curve gives better-than-even odds; these same tasks have a wage-weighted expected success of 68 per cent under Claude Opus 4.6 and 78 per cent under

Mythos Preview (early), not 100 per cent.

METR flags horizons beyond 16 hours as unreliable, which makes the integrated expected-success share the more stable aggregate. The p_{80} share is sensitive to the model's confidence interval because its horizon crosses the densest region of the duration distribution (Appendix E).

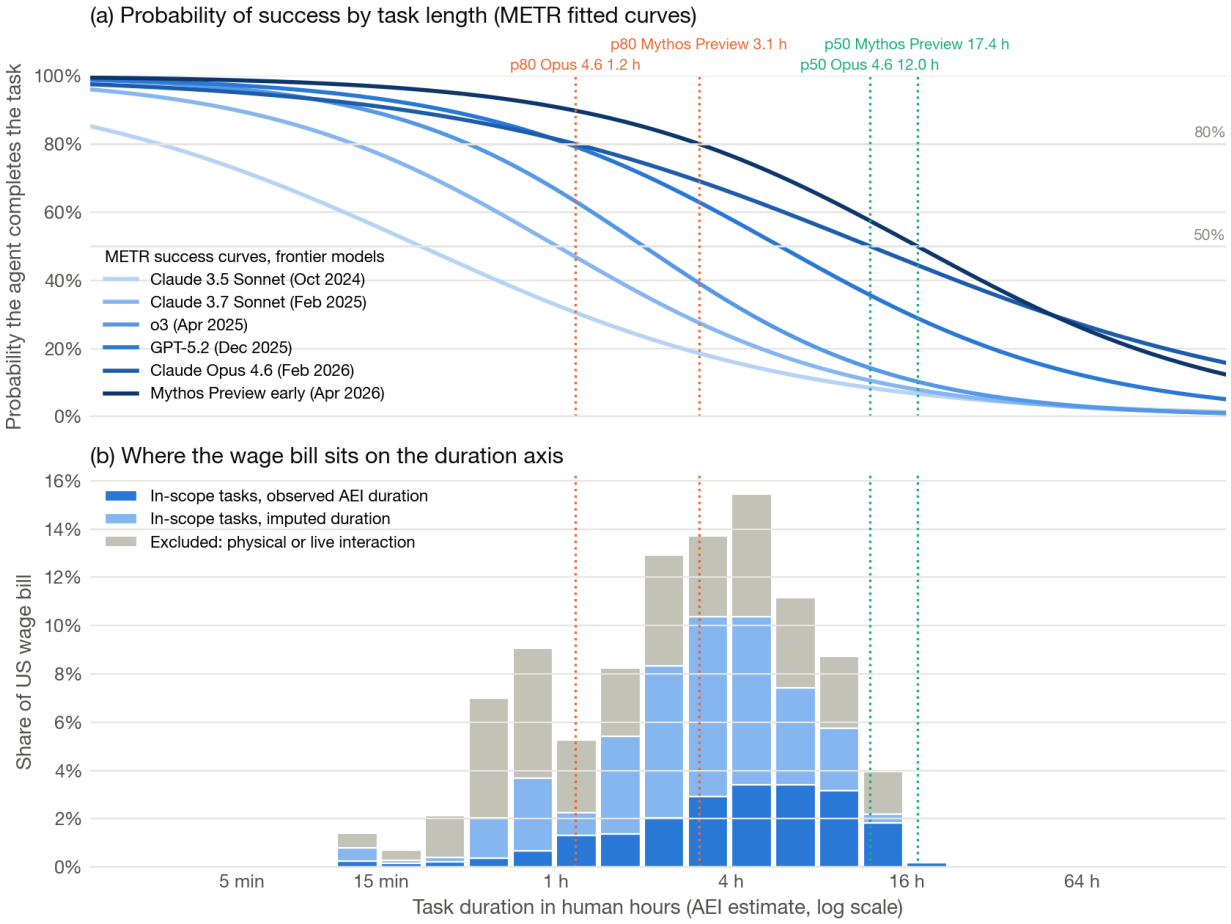


Figure 3: (a) METR's fitted success curves for six frontier models. (b) The US wage bill by task duration on the same axis, in half-$\log_2$ bins. Dotted lines: the 50 and 80 per cent horizons. The two 80 per cent horizons bracket 19 per cent of the wage bill.

4.2 Capability over time

Figure 4 traces the capability measures as the frontier advanced from 2023 to 2026. The expected-success share grew from 14 per cent under o1 to 46 per cent under Mythos Preview (early). The growth is similar under every scope rule of Section 2.4, a factor of 3, and larger under a stricter standard of success. On a flatter curve it grows far less (Section 5.4).

The expected-success share depends on the curve only through its 50 per cent horizon and its slope, and can be read off from those two numbers for any future METR estimate (Appendix Figure 14). At Mythos Preview (early)'s own slope, one more doubling of the horizon raises the share from 46 to 51 per cent: nearly all work in scope is already shorter than the 50 per cent horizon, so most of the remaining gap to the ceiling is the curve's failure rate on those tasks. At the median slope,

the share reaches 50 per cent at a horizon of about 28 hours, roughly 3 months beyond Mythos Preview (early) at the current doubling rate, and 55 per cent at about 79 hours (9 months). It then approaches its ceiling of 60 per cent, the in-scope share of the wage bill, slowly.

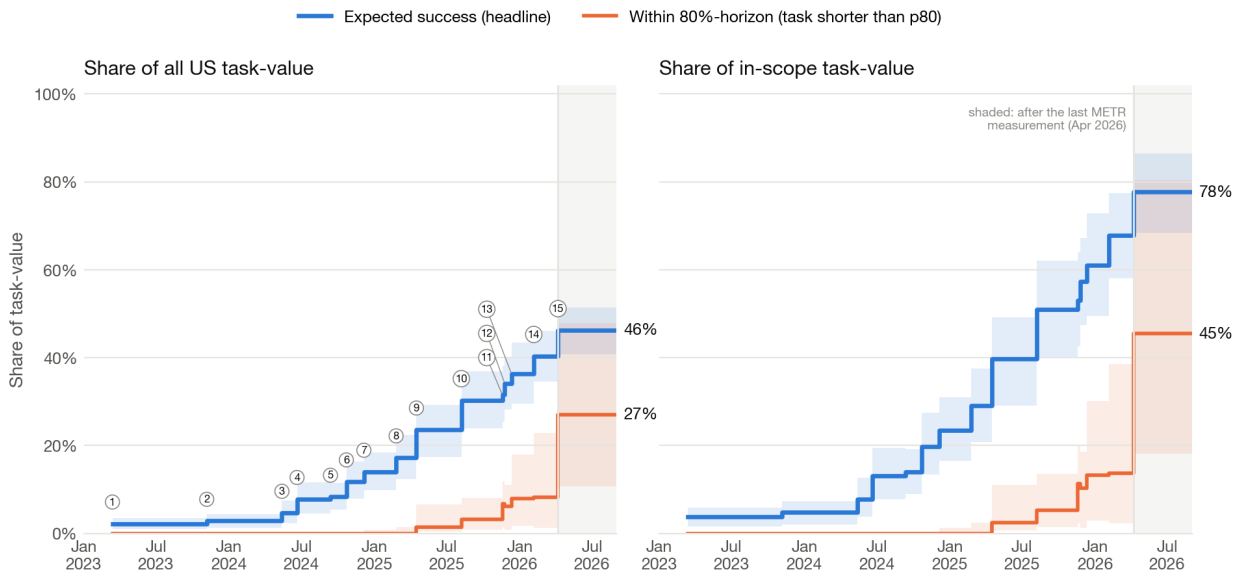


Figure 4: Share of the wage bill in tasks the frontier agent is expected to complete (blue) and in tasks shorter than the 80 per cent horizon (orange). Left: denominator is all tasks. Right: denominator is tasks in scope only. Frontier = longest 50 per cent horizon among models released by each date. Bands = 95 per cent intervals from the compound bootstrap.

4.3 Capability and observed use

Figure 5(a) compares capability with use across each AEI data window. Use counts conversations on Claude.ai and on the API; API data begin with the September 2025 window. Observed use within any single window ranges between 25 and 30 per cent of the wage bill, while the cumulative union across releases has grown to 49 per cent. The union is a generous measure of use, since it counts a task that has appeared in any window, however rarely; the expected-success share has risen from 14 to 46 per cent over the same windows. The cumulative measure depends on the definition of use: requiring a task to reach one in ten thousand conversations cuts the union to 24 per cent, and leaving out API traffic lowers it to 44 per cent (Appendix Table 15).

Figure 5(b) cross-tabulates the May 2026 wage bill. Observed use is not a subset of capability. At the 80 per cent horizon of Mythos Preview (early), 23 per cent of the wage bill sits beyond the capability frontier but still appears in Claude usage, which indicates that workers use AI on tasks the agent cannot yet complete reliably alone. 12 per cent of the wage bill falls inside the frontier but is unused. A further 12 per cent of the wage bill is in excluded tasks that nonetheless appear in Claude usage, most of it in tasks excluded for live interaction: users bring Claude the preparation and follow-up, not the meeting itself. Measured with the full curve rather than a threshold, 29 of the 46 points of expected success are in tasks that have appeared in Claude usage and 18 in tasks that have not; of the 49 per cent of the wage bill in use, 37 points are in scope.

The gap between capability and use is therefore not a simple adoption lag. The usage measure includes augmentation beyond the frontier and excludes unused capability within it. Short tasks

may also go unused because handing work to an agent has a fixed cost, in specifying the task and supplying its context (Appel et al., 2026), which is large relative to the time a short task saves. AEI data are also extensive-margin only: they record whether a task appears in conversations, not the intensity or the labour hours of the AI's contribution.

Task length is also associated with how AI is used. The AEI classifies each conversation by its pattern of collaboration. It counts as *automation* the directive conversations, in which the user hands Claude a task to complete, and feedback loops, in which the user only relays feedback such as error messages. Task iteration, learning and validation count as *augmentation* (Handa et al., 2025). Among the tasks in scope used on Claude.ai in May 2026, the automation share rises from 45 per cent in the tenth of tasks with the lowest success probability to 67 per cent in the highest (Figure 6). Comparing tasks within the same occupation, a success probability one standard deviation higher goes with an automation share 6.3 points higher.

The success probability falls with task length, which means that shorter tasks are more often handed over outright. That is what the METR curve predicts, since an agent is more likely to complete a short task correctly at the first attempt. It would also be expected if short tasks were simply easier to hand over in a single exchange, whatever the agent's ability, and the AEI's categories are defined partly by how much back-and-forth a conversation involves. Separating the two would require variation in capability at a given task length, such as the same task observed as the frontier moves.

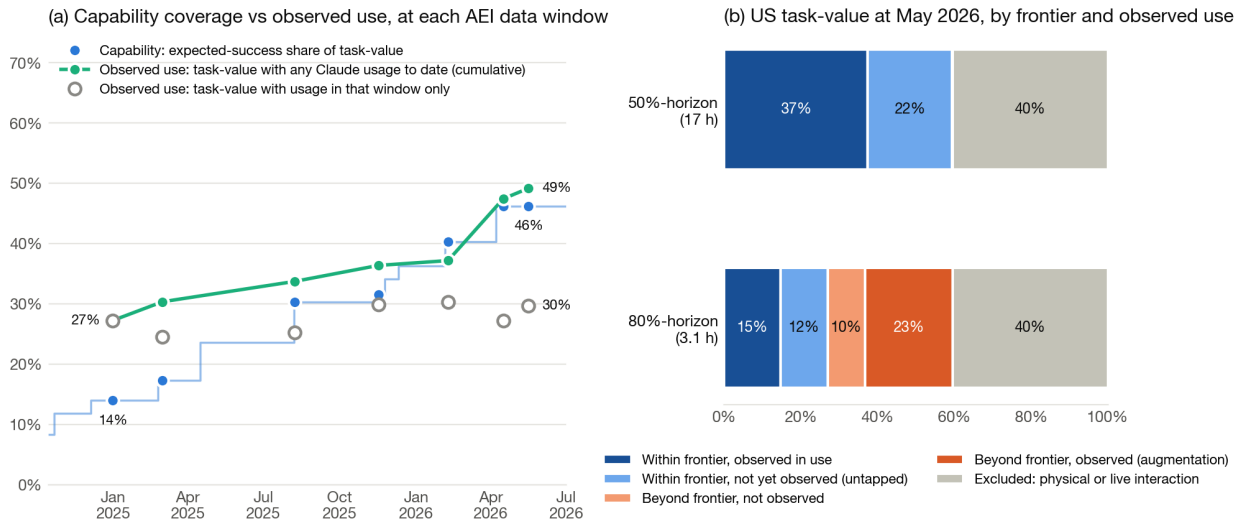


Figure 5: (a) At each AEI data window: expected-success share of the wage bill under the frontier as of the window's last day (blue); share of the wage bill in tasks with a published Claude.ai or API usage share in any window to date (green) and in that window only; API data begin with the September 2025 window (hollow grey). (b) The May 2026 wage bill split by whether a task is shorter than the horizon and whether it has appeared in any AEI release, for the 50 and 80 per cent horizons of Mythos Preview (early).

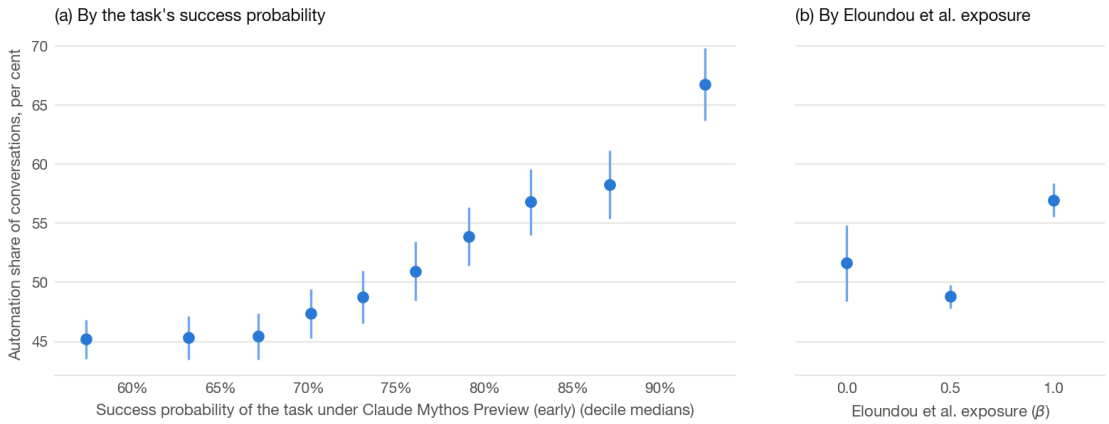


Figure 6: Automation share of Claude.ai conversations by task, May 2026: tasks in scope with a published share (2,378 tasks). (a) By decile of the task's success probability under Mythos Preview (early); points at decile medians. (b) By the Eloundou et al. (2024) exposure label. Bars: 95 per cent intervals. Automation: directive and feedback-loop conversations, as a share of conversations with a collaboration pattern (Handa et al., 2025). Evaluating the curve at durations imputed from similar tasks, leaving out a task's own conversations, gives a similar result.

4.4 Heterogeneity by occupation group

Figure 7 shows these measures across the 22 SOC major groups. Computer, engineering, office, business and science occupations have the highest expected-success shares. Cleaning, food service, construction and healthcare support rank lowest, because of the physical filter. Management, sales, education and the arts fall furthest when live interaction is excluded, because much of their work is directing, teaching or selling to people. Within the in-domain Computer and Mathematical occupations, the expected-success share is 63 per cent, and 82 per cent of the group's wage bill appears in cumulative Claude usage.

Appendix F varies the assumptions one at a time. Across the variants that keep the baseline denominator, and leaving aside the in-person rules of Section 2.4, the expected-success share ranges from 40 to 60 per cent, with the exclusion rule and the duration scaling producing the largest shifts.

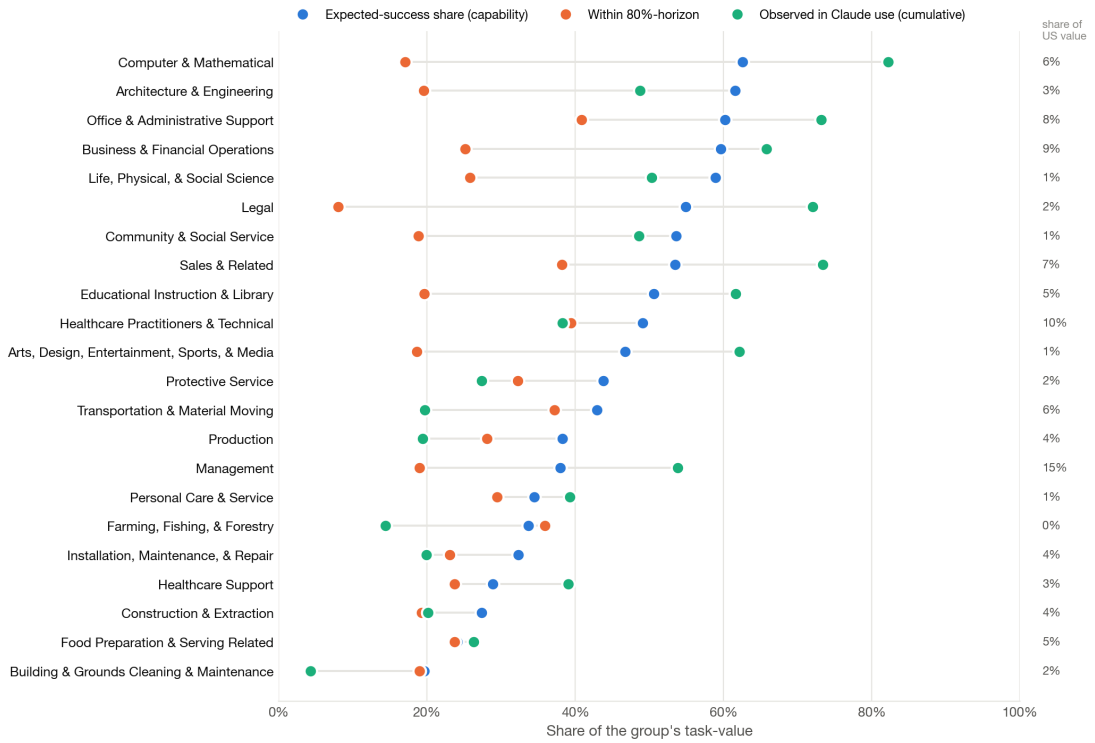


Figure 7: By SOC major group, under Mythos Preview (early): expected-success share (blue), share in tasks shorter than the 80 per cent horizon (orange), share with a published usage share in any AEI release (green). Denominator: the group's wage bill across all its tasks.

5 Discussion

5.1 From tasks to jobs

An occupation is a bundle of tasks, so coverage of tasks does not translate directly into displacement from jobs. For each 6-digit occupation I compute its own expected-success share and report the share of US employment above given thresholds (Table 16). I restrict this occupational analysis to jobs that can be performed remotely and without close physical contact (Section 2.4), assigning a probability of zero to the 66 per cent of US employment in occupations that require an in-person setting. Under Mythos Preview (early), 19 per cent of US employment is in occupations where at least half of the task value is expected to succeed, and 1 per cent where at least three quarters is. Figure 8 shows the same distribution for a stricter measure, the share of task value in tasks shorter than the 80 per cent horizon.

No occupation reaches 90 per cent of its task value under either model; the highest share is 85 per cent. The ceiling comes from the success curve rather than from excluded duties: the top occupation in Table 5 has no excluded tasks, and under Mythos Preview (early) even a one-hour task succeeds with probability 0.91. The ceiling therefore moves with the frontier. One more doubling of the horizon, about 4.2 months at the recent rate, would put 5 per cent of employment in occupations where at least three quarters of task value is expected to succeed, against 1 per cent now. Table 5 lists the occupations with the highest shares, which are records, transcription and clerical occupations, with some mapping and science occupations, whose tasks are short. Their durations are mostly imputed from related work activities, so the order within the top of the table

is not precise.

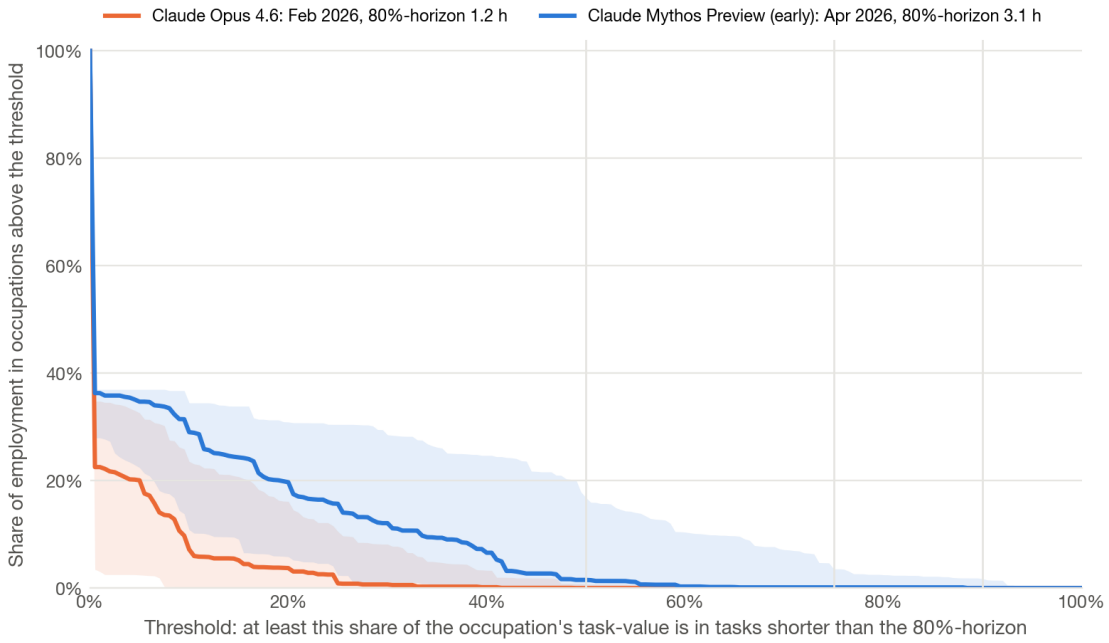


Figure 8: Share of employment in 6-digit occupations where at least the threshold share of the occupation's task value is in tasks shorter than the frontier model's 80 per cent horizon. Excluded tasks, and every task of an occupation that cannot be done remotely (Dingel and Neiman, 2020) or needs close physical contact, count against the occupation. Denominator: 153.4 million jobs. Bands: 95 per cent bootstrap intervals from the frontier model's fitted curve.

Aggregating task probabilities additively implies that tasks can be unbundled and automated one at a time. If the uncovered tasks are instead the core of the job, high task-level coverage may change occupational demand very little until the entire bundle is covered (Autor, 2015). Linear aggregation also overstates displacement when tasks are complements (Gans and Goldfarb, 2026), and whether automation raises or lowers the value of the remaining work depends on whether the automated tasks are the expert or the inexpert ones (Autor and Thompson, 2025). The shares are also measured at the grain at which work is delegated: an agent that completes the short pieces of a long task leaves the work of combining them to a person, and O*NET does not record that work. Capability is also not the same as comparative advantage. Delegating has a fixed cost of specifying the task and checking the result, so the short clerical tasks that top the occupational ranking may be the least worth handing over. With a weakest-link aggregator, in which an occupation is only as exposed as the lowest success probability among its more important tasks, 3 per cent of employment is above the 50 per cent threshold.

An expected-success share of 46 per cent therefore means partial capability across many occupations rather than the complete displacement of a few. Whether partial coverage turns into substitution or augmentation depends on the cost of delegating and on how firms reorganise work, which are beyond the scope of this paper.

Table 5: The ten occupations with the highest expected-success share of task value under Mythos Preview (early), alongside the five largest occupations by employment. Excluded tasks, and all tasks within occupations that cannot be done remotely (Dingel and Neiman, 2020) or require close physical contact, carry a success probability of zero. Appendix Table 20 replicates this ranking using the baseline task rule alone. Weakest link: the lowest success probability among the occupation's above-median-importance tasks (evaluating to zero if any such task is excluded). Median duration: wage-weighted baseline durations.

SOC	Occupation	Employment (thousands)	Expected success	Weakest link	Observed use	Median duration (h)
43-4021	Correspondence Clerks	4	85%	74%	85%	1.7
29-2072	Medical Records Specialists	195	82%	0%	71%	1.5
43-9041	Insurance Claims and Policy Processing Clerks	214	81%	69%	67%	3.4
31-9094	Medical Transcriptionists	42	80%	70%	84%	3.9
17-1021	Cartographers and Photogrammetrists	14	79%	71%	44%	2.8
17-3031	Surveying and Mapping Technicians	58	79%	56%	31%	2.5
43-9031	Desktop Publishers	3	79%	65%	78%	3.3
19-1011	Animal Scientists	3	79%	56%	42%	1.2
19-1013	Soil and Plant Scientists	16	78%	56%	22%	3.2
27-3092	Court Reporters and Simultaneous Captioners	13	78%	72%	30%	4.4
31-1120	Home Health and Personal Care Aides	4,306	0%	0%	50%	1.5
41-2031	Retail Salespersons	3,898	0%	0%	73%	2.1
35-3023	Fast Food and Counter Workers	3,854	0%	0%	22%	0.6
11-1021	General and Operations Managers	3,503	23%	0%	44%	7.0
29-1141	Registered Nurses	3,380	14%	0%	45%	2.9

5.2 Comparison with exposure measures

On the baseline filter, across 782 occupations, the rank correlation between the expected-success share and the value-weighted Eloundou et al. (2024) exposure score is 0.75. Most of this agreement comes from the excluded set, which both measures largely respect; computing both over tasks in scope only, the rank correlation is -0.41. The two measures answer different questions. The rubric asks whether an LLM could cut the time a person needs for a task by at least half, and does not consider how long the task is; the expected-success share asks whether an agent could complete the task alone, which becomes less likely as the task gets longer. Long analytical work, such as that of software developers (median task duration 8.1 hours), scores high on the first and lower on the second, while short clerical and front-line service work scores the other way. The long durations of software work partly reflect what users delegate: durations of coding-heavy tasks rose most when the AEI data began to include Cowork. Within occupations, the success probability is also more closely associated with the automation share of a task's Claude conversations than the exposure score is, with coefficients of 6.3 and 2.8 points per standard deviation in the same regression. Figure 13 in Appendix G plots the correlations.

5.3 Relation to scenario models

Table 6 sets this paper's estimates beside the values in the scenarios of Korinek et al. (2026). Their model lets AI affect only cognitive work, which they define as twelve SOC major groups that account for 0.624 of employment in the 2025 CPS, so for comparability the expected-success share I set against their m counts only tasks in those groups, divided by the whole wage bill as in their

definition. The expected-success share measures the work AI could perform outright, which their model calls automation, whether or not AI is yet used on it. In their notation it is closest to $\psi_t m_t$ when every instance is performed with AI ($d_t = 1$), and since any task an agent can complete alone is one that AI affects, it is also a lower bound on the affected mass measured as capability. Their mid-2026 anchor of 0.14 is built instead from Claude usage (Massenkoff and McCrory, 2026), and I reproduce it from the AEI files. It counts only tasks with substantial observed use, and it is low mainly for that reason: on this paper's wage-bill weights and in their twelve cognitive occupation groups, the same penetration measure gives 0.14, the share of the wage bill in tasks that appeared in Claude usage in May 2026 is 0.26, and the share in tasks that appeared in any window is 0.42. The expected-success share under Mythos Preview (early) is 0.36; the 46 per cent of Section 4 also counts the occupation groups that their model leaves unaffected. The capability measure is therefore not larger than every measure of use; what distinguishes it is that it measures what AI can do rather than how far its use has spread, and that it moves with the frontier. If METR's curves carry over to the rest of the economy, this lower bound already lies between the 0.3 and 0.5 that their substantial and extreme scenarios assume for m in 2030, as it does between the values of ψm that those two scenarios imply.

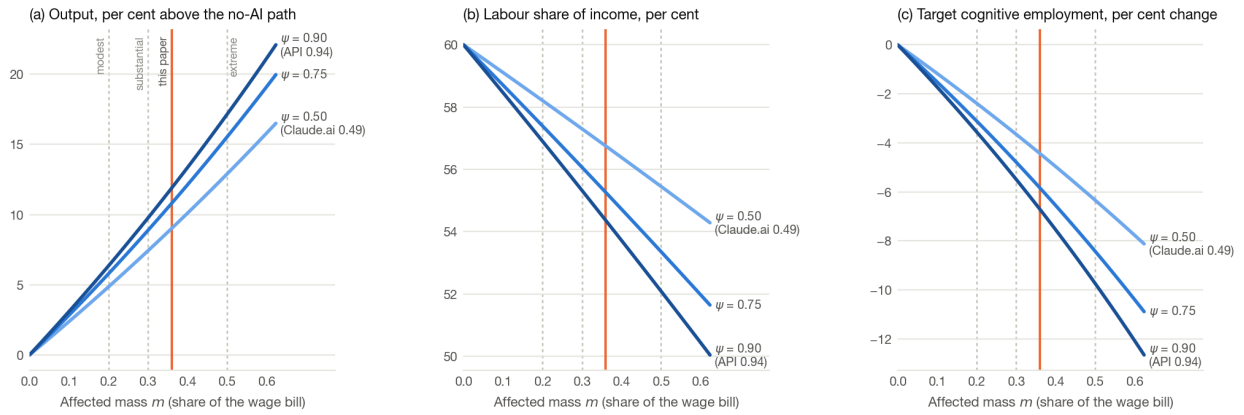


Figure 9: Static targets of the model of Korinek et al. (2026) as the affected mass m varies, for their three automation shares ψ. Diffusion (0.4), the gain per instance (0.45) and reinstatement (0.25) are at their substantial 2030 values; capital supply elasticity 3. Grey dotted lines: their 2030 values of m. Orange: this paper's expected-success share under Mythos Preview (early) in the twelve occupation groups they classify as cognitive, as a share of the whole wage bill (0.36), a lower bound on m; counting only occupations that can be done remotely and without close contact it is 0.23. Labels in brackets: the AEI's measured automation shares, June 2026. (c) The level of employment cognitive occupations adjust toward, not employment in 2030.

Figure 9 solves the static part of their model, their Proposition 1, which gives the labour share, output and the employment each occupation group adjusts toward, before the dynamics of job search and of the stock of ideas.[1] The affected mass and the diffusion share enter the model only as a product, so the affected mass moves wages and employment only in proportion to diffusion. With capability already above their substantial value, the spread of their 2030 outcomes therefore depends mainly on diffusion, the gain per task and the automation share. In their modest scenario the labour share falls by 0.6 points by 2030; with the affected mass set to the measured value and

[1] At their parameter values it reproduces their published labour share for 2030 to within 0.1 points in the modest and substantial scenarios and 0.6 points in the extreme scenario, and the net return to capital to within 0.1 points in the first two (their Table 3). Output and cognitive employment are targets. They run ahead of the published paths, which include the time workers take to move between jobs.

nothing else changed, it falls by 1.1 points, against 15 points in the extreme scenario.

Across tasks, the automation share is higher where the success probability is higher (Section 4.3), which fits their choice to pair a larger affected mass with a higher automation share. Its level depends more on how AI is deployed than on what it can do: about half of Claude.ai conversations are automation, against nine tenths of API traffic.

Table 6: The parameters of Korinek et al. (2026) beside this paper's empirical measures. Their values are published figures (their Table 1): the mid-2026 anchor, where provided, and the modest / substantial / extreme values assumed for 2030. Shares are expressed as a fraction of the wage bill; "cognitive occupations" refer to their twelve specified SOC major groups. Expected success is evaluated using METR's success concept under Mythos Preview (early). The AEI automation share is the usage-weighted share of conversations classified as automation (directive or feedback-loop use) in the June 2026 release.

Dial	Korinek et al. (2026): mid-2026 anchor; 2030 modest / substantial / extreme	This paper
Affected mass m: share of tasks AI can affect	0.14 (Claude usage); 0.2 / 0.3 / 0.5; ceiling 0.624 (all cognitive work)	Expected success in cognitive occupations: 0.36 (0.23 in remote jobs without close contact). Claude use (Claude.ai and API), May 2026: 0.30. Ceiling: 0.60 (all in-scope work). Expected success is a lower bound on m, and closest to ψm with every instance done by AI
Diffusion share d: instances of affected tasks done with AI	0.10 (Census firm survey); 0.2 / 0.4 / 0.6	not measured
Automation share ψ: AI-performed instances automated	0.50 / 0.75 / 0.90, held fixed	AEI, June 2026: 0.49 Claude.ai, 0.94 API. Higher for tasks with a higher success probability, across tasks (Section 4.3)
Gain per AI-performed instance a (log points)	0.30 / 0.45 / 0.80	not measured
Reinstatement ρ: new tasks per automated task	0.5 / 0.25 / 0	not measured

5.4 A flatter success curve

On the flatter curve of Mertens et al. (2026), the expected-success share starts higher and grows far less. Their models were released between June 2023 and August 2025, ending with GPT-5, so the comparison stops there; I do not extrapolate their curve to Mythos Preview (early). From o1 at the start of 2025 to GPT-5, their release-date trend gives a share that rises from 37 to 41 per cent of the wage bill, while METR's curves give 14 and 30 per cent: growth by a factor of 1.1 rather than 2.2. The threshold shares would say little. At GPT-5's release, most of the wage bill in scope would sit at success rates between 66 and 73 per cent, against 31 to 76 per cent on METR's curve, all of it shorter than their 50 per cent horizon and almost none of it shorter than their 80 per cent horizon. Their slope also varies across occupations, from 0.39 per tenfold increase in length in personal care to none in legal work, which a single curve ignores. Their curve measures single responses that workers judge usable without edits, on tasks whose length the same workers estimate and with all context supplied. They note that supplying context in practice is costly, which is one reading of the gap between capability and use in Section 4.3.

5.5 Limitations

Four limitations matter most. The first is the duration mismatch. METR measures human time for complete, well-specified tasks performed by contractors, whereas the AEI estimates time for the modular pieces of work users delegate in chat. Delegated pieces are short by selection, so AEI durations may understate the length of employer-defined tasks and bias the capability shares upward. Doubling every duration lowers the expected-success share under Mythos Preview (early) to 40 per cent (Appendix F). METR's human times also come from low-context contractors, who take 5 to 18 times longer than repository maintainers on the same issues (Kwa et al., 2025). If AEI durations instead reflect experienced workers in context, the capability share falls to 18–31 per cent (Table 18). Claude's time estimates correlate with actual task times on software tickets about as well as developers' own estimates (Tamkin and McCrory, 2025), but only 26 per cent of the wage bill has an observed duration. Hold-out checks show the imputation is unbiased at the level of detailed work activities (Appendix Table 14), but not whether it carries over to tasks that are never delegated.

Second, the success curve is extrapolated beyond software. Within METR's suite, the curve overstates success in the one-to-four-hour band; reweighting the residuals suggests this inflates the headline share by about 2 points, which is within sampling noise (Appendix Table 9). Outside the suite, success curves derived from AEI usage, GDPval and worker evaluations across O*NET tasks are all flatter. The worker evaluations come closest to autonomous completion, but of self-contained tasks answered in a single response rather than work an agent carries out step by step (Sections 3 and 5.4). The top of the frontier is also poorly measured, which makes the 80 per cent threshold share uncertain (Appendix E).

Third, the level depends on what counts as success. METR scores well-specified tasks automatically. When maintainers judge whether they would merge the code instead, METR's horizons are about 7 times shorter (Whitfill et al., 2026), which lowers the expected-success share to 27 per cent. The gap is wider still for freelance work: on the Remote Labor Index, METR's curves predict a 30 to 51 per cent completion rate, yet clients accepted only 1.2 to 4.2 per cent of the AI deliverables (Mazeika et al., 2025). I do not convert that gap into a share, because the benchmark reports one rate per model and 69 per cent of its projects are in video, audio, design, CAD and game development (Appendix H). The gaps between standards are far larger than the sampling interval in Table 4. The exclusion of live interaction may also go too far: voice agents already answer telephones, and some teaching and supervision can be done in writing. Keeping selling in scope gives 47 per cent, and the physical filter alone 60 per cent.

Fourth, the measure gives a task the agent completes with probability 0.7 credit for 70 per cent of its value, with no allowance for the time a worker spends checking the output. If checking an agent's work takes a quarter of the task's original human time, the net expected-success share falls to 31 per cent (Appendix K). Only 8.0 per cent of the wage bill sits in tasks the curve expects to succeed nine times in ten, and some tasks need higher reliability still before they are worth delegating (Kwa, 2026). Experienced developers in METR's own trial were slower with AI than without it (Becker et al., 2025).

If METR's curves carry over, the capability frontier covers in expectation just under half of the US wage bill, up from 14 per cent at the start of 2025. Its level depends on which tasks count as within the scope of an agent, which the data cannot yet settle; its growth, a factor of 3 since the start of 2025, holds under every definition I consider. Further gains depend more on how reliably agents complete short tasks than on how long a task they can attempt. For now the coverage is broad but

partial: the frontier reaches parts of most jobs that can be done remotely and the whole of few. For macroeconomic models of AI, the measure gives a capability-based value for the share of work AI could automate. At that value, the spread of outcomes in the scenarios of Korinek et al. (2026) depends mainly on how far AI use spreads, what it gains per task and how much of it is automation, and the first two have yet to be measured. Refining the estimates requires human completion times for tasks defined at the level of the occupation, and success curves measured outside software under a stated standard of success, such as those of Mertens et al. (2026).

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A Data Construction

The estimates in this paper are constructed from a single task-level dataset. Every aggregate reported in the text derives from an analysis file with one row per O*NET task, which records the task's wage weight, observed or imputed AEI duration, physical classification, Eloundou et al. (2024) exposure label, usage flags and fitted success probabilities under the frontier models.

- **METR time horizons:** Data are downloaded from `metr.org/time-horizons` (`benchmark_results_1_1.yaml` and `task_results_1_1.yaml`, updated 8 May 2026). The curve refits follow METR's public methodology (`github.com/METR/eval-analysis-public`): a weighted logistic regression of run success on $\log_2$ human minutes, with weights equal to the family weight divided by runs per task, normalised to one per model, using 10^{-5} regularisation. Standard errors and intervals are generated via 500 bootstrap draws over family, task, and run. The frontier at date d is defined as the model with the longest 50 per cent horizon released on or before d, and an AEI data window is paired with the frontier as of its last day. For the robust variants (Appendix F), the curves are refit using a probit link, logistic functions with asymptotes fixed at 0.85, 0.90, or 0.95, HCAST tasks only, and a leave-one-family-out procedure.
- **Anthropic Economic Index (AEI):** Sourced from `huggingface.co/datasets/Anthropic/EconomicIndex`. Usage shares are extracted from seven Claude.ai data windows spanning December 2024 to May 2026, and from the five API windows published from September 2025. Per-task human time (`human_only_time_mean`, which averages the estimated hours for a human to complete the delegated work without AI) uses the June 2026 release as the baseline, as it uniquely provides exact O*NET task IDs. Releases prior to June 2026 identify tasks by text string; I recover 87 to 91 per cent of usage via exact string matching and drop the residual (mostly retired O*NET occupations). Missing durations are imputed using the simple mean over a task's detailed work activities of the geometric mean of observed durations in each activity (36 per cent of the wage bill). If no activity has an observed peer, the imputation backs off sequentially to the intermediate (22 per cent) and generalized work activity (15 per cent), and finally to the global geometric mean (0.8 per cent). Tasks map to 1.27 detailed work activities on average.
- **O*NET 30.2:** Task statements, ratings (importance, relevance, frequency), and the work-activity hierarchy are merged to construct the wage weights. These weights use O*NET's core tasks (relevance ≥ 67 per cent and mean importance ≥ 3.0) with positive relevance; specialties are averaged into the 6-digit SOC code, and 29 occupations missing ratings receive equal weights. Physical tasks are defined by mapping to the following generalized work activities: 4.A.3.a.1–4 (performing general physical activities, handling and moving objects, controlling machines and processes, operating vehicles and equipment), 4.A.3.b.4–5 (repairing and maintaining equipment) and 4.A.4.a.5 (assisting and caring for others). Live-interaction tasks are defined by 4.A.4.a.4, 4.A.4.a.6, 4.A.4.a.7 and 4.A.4.a.8 (relationships, selling or influencing, negotiating, working with the public), 4.A.4.b.2 to 4.A.4.b.5 (building teams, teaching, directing subordinates, coaching) and 4.A.4.c.2 (staffing). The Dingel–Neiman rule uses the Work Context and Work Activities files of the same release (Appendix J).
- **OEWS May 2025 (National):** Provides employment and mean annual wage by detailed occupation. Hourly wages are annualised by OEWS at 2,080 hours. Twelve OEWS hybrid codes are mapped explicitly. Occupations lacking O*NET task data (accounting for 1.3 per cent of employment) are dropped.
- **Eloundou et al. (2024) labels:** Sourced from `github.com/openai/GPTs-are-GPTs` and used solely to define the alternative excluded set and for the rank correlation comparison in Section 5.2.
- **External benchmarks:** METR's cross-domain horizons are taken from `github.com/METR/`

`cross-domain-horizon` (pinned commit). Published figures from GDPval (win rates by task length and the distribution of task times; Patwardhan et al., 2025), METR's studies of maintainer review and of contractor and maintainer times (Whitfill et al., 2026; Kwa et al., 2025), the Remote Labor Index (Mazeika et al., 2025), the AEI January 2026 report (Appel et al., 2026), Korinek et al. (2026) and the fitted curves of Mertens et al. (2026) are recorded, with page references and access dates, in the replication files. They are cited parameters rather than data, and every table that uses one marks it. I read their curves with human time in minutes.

An interactive version of the task-level data, in which all 18,796 tasks are drawn as individual marks arranged by occupation group and duration against the frontier, is available at `https://aitasks-ecru.vercel.app/`.

B METR Task Suite and Refit Verification

Table 7 gives the composition of METR's Time Horizon 1.1 suite by source. Table 8 reports METR's published 50 and 80 per cent horizons alongside the refit, and the fitted slope per doubling of task length, for all models with fitted curves. "Frontier at release" is METR's `is_sota` indicator; these models are the frontier steps in Figure 4. Table 9 reports the fit's residuals by duration bin.

Table 7: Composition of METR's Time Horizon 1.1 task suite. Weight = share of METR's diversity-adjusted weight in the Mythos Preview (early) fit, where each task in a family of n_f tasks carries weight $1/\sqrt{n_f}$, so a family's total weight is proportional to $\sqrt{n_f}$. The band from half an hour to eight hours holds 40 per cent of the weight on 72 tasks.

Source	Content	Tasks	Families	Human time: median, max	Weight
HCAST	software engineering, machine learning, cybersecurity, and computer-based research tasks	157	69	44 min, 30.0 h	81%
RE-Bench	machine-learning research engineering tasks	5	5	8.0 h, 8.2 h	4%
SWAA	seconds-long software actions (code completion, file selection, request routing)	66	5	5 s, 43 s	15%

Table 8: METR's published 50 and 80 per cent horizons and my refit from the per-task results, all models with fitted curves. Slope = the refit coefficient on $\log_2$ human minutes, sign reversed: the fall in the log-odds of success per doubling of task length. "Frontier at release" is METR's `is_sota` flag; these models are the numbered steps in Figure 4, in date order from GPT-4 (March 2023). Mythos Preview (early) is a pre-release checkpoint measured on 1,149 runs.

Model	Released	p50 METR (min)	p50 refit (min)	p80 METR (min)	p80 refit (min)	Slope	Frontier at release
GPT-4 0314	2023-03-14	4.0	4.0	0.9	0.9	0.64	yes
GPT-4 1106	2023-11-06	4.0	4.0	0.8	0.8	0.59	yes
Claude 3 Opus	2024-03-04	4.0	4.0	0.6	0.6	0.53	
GPT-4 Turbo	2024-04-09	3.7	3.7	0.9	0.9	0.69	
GPT-4o	2024-05-13	7.0	7.0	1.3	1.3	0.56	yes
Claude 3.5 Sonnet (Old)	2024-06-20	11.4	11.4	1.7	1.7	0.50	yes
o1-preview	2024-09-12	20.3	20.3	4.4	4.4	0.63	yes
Claude 3.5 Sonnet (New)	2024-10-22	20.5	20.5	2.6	2.6	0.46	yes
o1	2024-12-05	38.8	38.8	7.1	7.1	0.57	yes
Claude 3.7 Sonnet	2025-02-24	60.4	60.4	12.1	12.1	0.60	yes
o3	2025-04-16	119.7	119.7	30.0	30.0	0.69	yes
Claude 4 Opus	2025-05-22	100.4	100.4	20.4	20.4	0.60	
Claude 4.1 Opus	2025-08-05	100.5	100.5	23.5	23.5	0.66	
GPT-5	2025-08-07	203.0	203.0	38.3	38.3	0.58	yes
Gemini 3 Pro	2025-11-18	224.3	224.3	54.1	54.1	0.68	yes
GPT-5.1-Codex-Max	2025-11-19	223.7	223.7	50.6	50.6	0.65	
Claude Opus 4.5	2025-11-24	293.0	293.0	49.4	49.3	0.54	yes
GPT-5.2	2025-12-11	352.2	352.3	66.0	66.0	0.57	yes
GPT-5.3-Codex	2026-02-05	349.5	349.5	54.7	54.7	0.52	
Claude Opus 4.6	2026-02-05	718.8	718.8	69.9	69.9	0.41	yes
Gemini 3.1 Pro	2026-02-19	384.1	384.1	89.8	89.8	0.66	
GPT-5.4	2026-03-05	341.7	341.7	53.9	53.9	0.52	
Claude Mythos Preview (early)	2026-04-07	1044.8	1044.7	185.9	185.9	0.56	yes

Table 9: Observed minus fitted success by duration bin within METR's suite, Mythos Preview (early). Observed and fitted are diversity-weighted means over the tasks in the bin; difference in percentage points. Last column: share of the total wage bill in tasks in scope whose AEI duration falls in the bin. Reweighting the differences by the last column gives the change in the expected-success share that would follow if the suite's residual pattern held in the economy: about −2 points for Mythos Preview (early) and −1 for Claude Opus 4.6.

	METR suite		Mythos Preview (early)			In-scope wage bill
Human time	Tasks	Weight	Observed success	Fitted	Difference	in this bin
< 1 min	67	15%	99%	100%	-0.5	0%
1-15 min	52	24%	99%	98%	+0.3	1%
15-60 min	36	17%	98%	94%	+3.3	6%
1-4 h	18	9%	76%	86%	-9.9	26%
4-16 h	50	31%	68%	67%	+1.3	26%
> 16 h	5	3%	45%	47%	-1.4	0%

C Task Examples and Classification

Table 10 lists the twelve METR task families with the largest weight in the fit; the family names are METR's identifiers, and Rein et al. (2025) and Wijk et al. (2024) describe the tasks. Table 11 shows, for eight SOC major groups, the O*NET task with the largest wage bill in the group, with its duration on the baseline axis. Figure 10 shows every O*NET task of one occupation, Data Entry Keyers, with the exclusion rule applied.

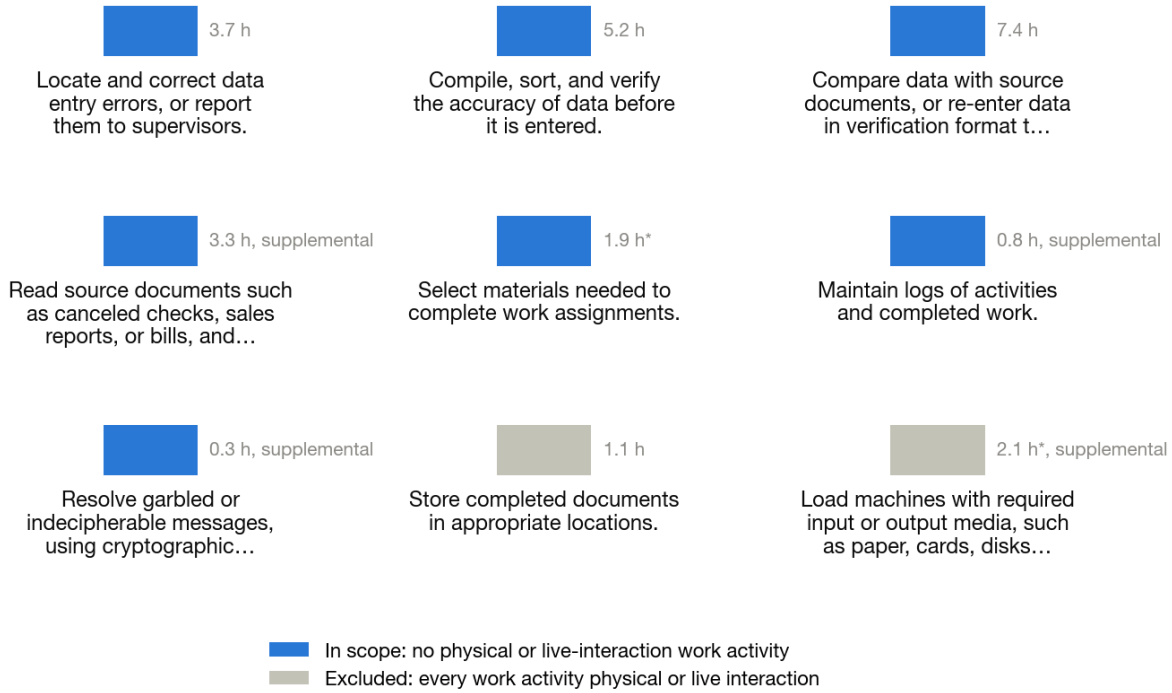


Figure 10: The 9 O*NET tasks of Data Entry Keyers (O*NET-SOC 43-9021.00), two of which appear in Table 1. Each tile is one task statement, shortened. Dark blue: no physical or live-interaction work activity. Grey: every work activity physical or live interaction, so excluded. Excluded tasks never count as within the frontier. Grey text: the task's AEI human duration on the baseline axis, the estimated human time for the work delegated in a conversation, averaged over conversations; * = imputed. Only the 5 core tasks carry wage weight; supplemental tasks carry none. Tiles are ordered by classification and then by O*NET importance.

Table 10: The twelve METR task families with the largest diversity-adjusted weight in the Mythos Preview (early) fit.

Task family	Source	Tasks	Human time, shortest to longest	Weight
`arithmetic`	SWAA	20	1 s to 43 s	3.7%
`file_selection`	SWAA	20	2 s to 6 s	3.7%
`alert_triage`	SWAA	10	4 s to 10 s	2.6%
`env_scientist`	HCAST	9	44 min to 11.1 h	2.5%
`request_routing`	SWAA	9	8 s to 18 s	2.5%
`pico_ctf`	HCAST	7	2 min to 1.1 h	2.2%
`data_cleaning_arjun`	HCAST	7	6 min to 6 min	2.2%
`code_completion`	SWAA	7	3 s to 6 s	2.2%
`palisade_crackme`	HCAST	6	21 min to 8.0 h	2.0%
`blackbox`	HCAST	6	10 min to 1.1 h	2.0%
`password_check`	HCAST	6	57 s to 32 min	2.0%
`hypothesis_testing`	HCAST	6	9 min to 49 min	2.0%

Table 11: The O*NET task with the largest wage bill in each of eight SOC major groups. Hours = AEI human duration on the baseline axis: the estimated time for a person to do, without AI, the piece of the task that users delegate in one conversation, averaged over conversations mapped to the task; * = imputed. The wage weight attaches to the whole statement. Excluded = at least one of the task's work activities is physical or live interaction, so the task never counts as within the frontier.

Occupation	Task (O*NET statement)	Hours	Excluded
General and Operations Managers	Review financial statements, sales or activity reports, or other performance data to measure productivity or goal...	7.0	
Software Developers	Analyze user needs and software requirements to determine feasibility of design within time and cost constraints.	11.3	
Lawyers	Interpret laws, rulings and regulations for individuals and businesses.	4.3	
Registered Nurses	Record patients' medical information and vital signs.	2.0*	
Retail Salespersons	Greet customers and ascertain what each customer wants or needs.	0.5	yes
Customer Service Representatives	Confer with customers by telephone or in person to provide information about products or services, take or enter orders...	0.8	
Construction Laborers	Tend pumps, compressors, or generators to provide power for tools, machinery, or equipment or to heat or move materials...	2.1*	yes
Laborers and Freight, Stock, and Material Movers, Hand	Maintain equipment storage areas to ensure that inventory is protected.	0.6*	

D Variation within METR's Suite

Figure 11 plots, for the two frontier models, each METR task's observed success rate against its human time, with METR's pooled curve and curves fitted within each task source. Table 12 gives the horizons from each fit and the range of horizons obtained by dropping any one task family and refitting. The R^2 in the panel titles is the weighted R^2 of task success rates on the pooled curve; the corresponding ceiling if the curve were exactly right, given the number of runs per task, is 0.61 for Claude Opus 4.6 and 0.57 for Mythos Preview (early). Across all 23 models with fitted curves the highest weighted R^2 is 0.68 (GPT-4 0314, ceiling 0.89); the frontier models sit low because most of their tasks are solved nearly every time and the remaining failures are concentrated in particular tasks rather than at particular lengths. The RE-Bench fits are not shown because five tasks do not identify a slope. The SWAA-only horizons are shown but should not be read as measurements: the longest SWAA task takes 43 seconds, so both horizons for Mythos Preview (early) and the 50 per cent horizon for Claude Opus 4.6 lie beyond the subsuite's data, and one percentage point of observed success separates a one-minute from a sixteen-minute fitted horizon.

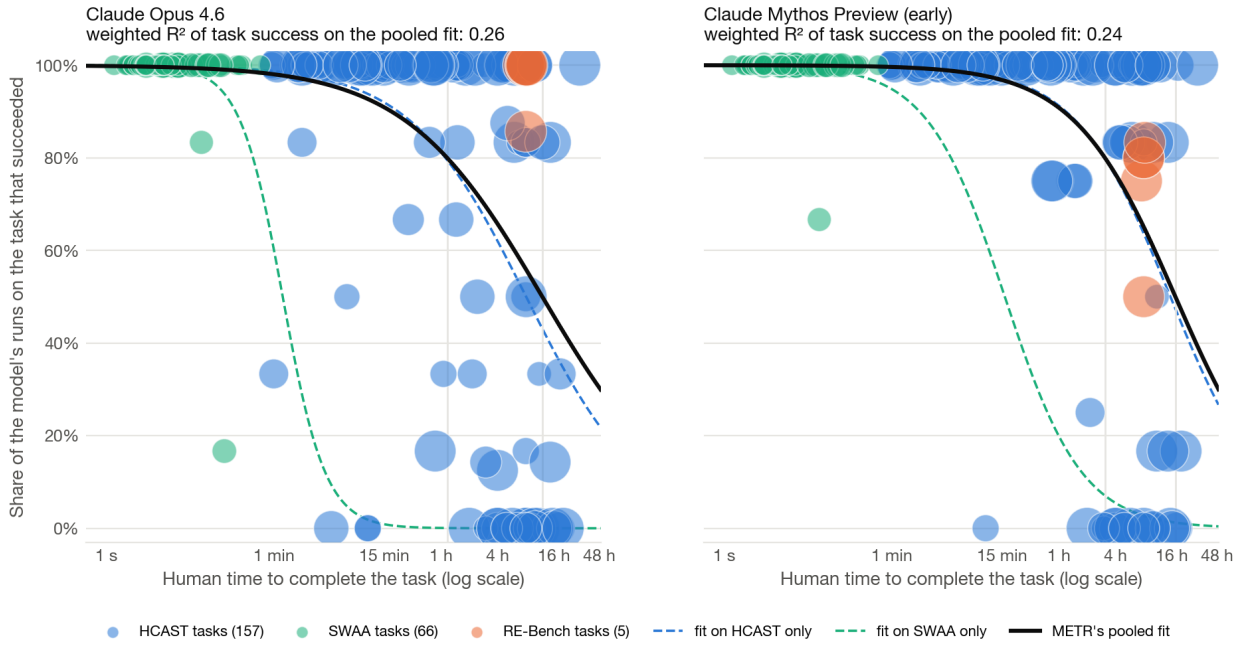


Figure 11: Each point is one METR task; area is proportional to METR's diversity weight. Black: METR's pooled fit. Dashed: logistic fits within one task source. Grey lines: the pooled 50 and 80 per cent horizons. Left: Claude Opus 4.6. Right: Mythos Preview (early).

Table 12: Horizons from METR's pooled fit, from fits within one task source, and the range when any one of the 79 task families is dropped. Success = diversity-weighted share of runs that succeeded. †: horizon lies beyond the longest task in the subsuite (an extrapolation, not a reading within the data).

Model	Fit	Tasks	Weight	Success	50% horizon	80% horizon
Claude Opus 4.6	pooled (METR's fit)	228	100%	79%	12.0 h	1.2 h
	HCAST tasks only	157	81%	74%	8.1 h	1.1 h
	SWAA tasks only	66	15%	98%	1 min†	36 s
	RE-Bench tasks only	5	4%	97%	n/a	n/a
	drop any one family				10.3 h to 14.2 h	1.1 h to 1.4 h
Mythos Preview (early)	pooled (METR's fit)	228	100%	85%	17.4 h	3.1 h
	HCAST tasks only	157	81%	83%	15.4 h	3.1 h
	SWAA tasks only	66	15%	99%	16 min†	4 min†
	RE-Bench tasks only	5	4%	74%	n/a	n/a
	drop any one family				14.8 h to 20.9 h	2.9 h to 3.6 h

E Decomposing the April 2026 Capability Step

The 80 per cent threshold share in Figure 4 jumps from 8 to 27 per cent when the frontier passes from Claude Opus 4.6 to Mythos Preview (early). The step is a faithful reading of METR's published fits, but three things make it as large as it is. First, the frontier only moves when a model beats the best 50 per cent horizon to date. METR measured two other models between these two: Gemini 3.1 Pro (19 February 2026; 50 per cent horizon 6.4 hours, 80 per cent horizon 1.5 hours) and GPT-5.4 (5 March 2026; 50 per cent horizon 5.7 hours, 80 per cent horizon 0.9 hours). Neither beat the 50 per cent horizon of Claude Opus 4.6, so neither appears in the figure, although the 80 per cent horizon of Gemini 3.1 Pro was the longer of the two and would have put 10 per cent of the wage

bill below it; its expected-success share, 38 per cent, is below Claude Opus 4.6's, so the frontier rule and the share agree on the ranking. Second, Claude Opus 4.6 has an unusually flat success curve, 0.41 per doubling against 0.56 for Mythos Preview (early). Its 50 per cent horizon is 10 times its 80 per cent horizon, against 4 to 6 times for the other frontier models since 2025, so its 80 per cent horizon barely moved from that of GPT-5.2 (1.1 to 1.2 hours) while its 50 per cent horizon doubled. Much of the 6-point rise in the expected-success share between the two models is therefore the slope steepening in the half-hour-to-eight-hour band rather than the horizon lengthening. Third, the 80 per cent threshold share counts tasks shorter than a threshold, and the threshold moved from 1.2 to 3.1 hours, across the part of the duration axis where the wage bill is: 19 per cent of the wage bill is in tasks in scope with durations between those two values (Figure 3(b)), and 26 per cent is in tasks in scope between one and four hours. The expected-success share integrates the curve instead of counting a threshold, and moves 6 points on the same step.

The threshold is also poorly determined at the top. The 80 per cent horizon of Mythos Preview (early) has a 95 per cent interval of 1.6 to 6.6 hours, which puts the share anywhere between 11 and 48 per cent (the orange band in Figure 4). Under Claude Opus 4.6 the interval is 1 to 23 per cent. The level of the orange line in 2026 is therefore uncertain by tens of points either way, and the size of the step says as much about where the wage bill sits on the duration axis as about the model.

F Robustness and Sensitivity Analysis

Table 13 changes one assumption at a time, with Mythos Preview (early) as the frontier, in five groups. Across the variants that keep the baseline denominator, other than the in-person rules, the expected-success share ranges from 40 to 60 per cent. The duration bridge and the exclusion rule matter most. Doubling every duration gives 40 per cent and cuts the 80 per cent threshold share to 11 per cent; halving them raises that share to 46 per cent. Inflating imputed durations alone, which targets the selection of the observed sample rather than the level of the clock, gives 44, 42 and 40 per cent at factors of 1.5, 2 and 3. Using the earlier AEI releases for durations raises the share by 3 points. On the curve side, refitting on HCAST tasks only, dropping any one family, switching to a probit link or fixing the asymptote at 0.85 to 0.95 all leave the share within two points of the baseline, although the probit and low-asymptote fits move the 80 per cent threshold share to 23 and 20 per cent. Task weights within occupations barely matter, including the hours-implied weight. Switching off O*NET's core filter and weighting every rated task by importance times relevance moves 16 per cent of the wage bill onto supplemental tasks but gives 46 per cent; restricting that to tasks with importance above 2.5 and relevance above 50 per cent gives 46 per cent. Dropping wages and weighting by employment gives 44 per cent. The lenient rule gives 52 per cent, the physical filter alone 60 per cent, keeping selling in scope 47 per cent, the Eloundou labels 49 per cent, and crediting mixed tasks' in-scope share at a proportionally shorter duration 48 to 51 per cent. The in-person rules of Section 2.4 give 44 per cent at the task level and 26 per cent with the Dingel–Neiman bound. The two observed-use variants leave the expected-success columns unchanged by construction; what they move is the use column, from 49 to 24 per cent under a share threshold and to 44 per cent without API traffic. The last two rows restrict the universe to tasks with an observed duration and are not commensurable with the rest; they are the source of the 62 per cent that appears if the whole column is read as one range.

Table 13: One assumption changed at a time; frontier = Mythos Preview (early). Denominator: all tasks, except in the last two rows, where it is the observed-duration tasks. Curve variants refit the same model's runs another way (Appendix A). θ: share of a mixed task's value credited as in scope, evaluated at duration θD_t. The in-person rows are described in Appendix J.

Variant (one change from baseline)	Expected success	< p80	Observed, cumulative
Duration bridge			
Baseline	46%	27%	49%
Durations from the Nov 2025 and Feb 2026 AEI releases	49%	40%	49%
All durations halved	51%	46%	49%
All durations doubled	40%	11%	49%
Imputed durations only ×1.5	44%	18%	49%
Imputed durations only ×2	42%	15%	49%
Imputed durations only ×3	40%	13%	49%
Success curve (same model, refit)			
Curve fitted on HCAST tasks only	46%	26%	49%
Drop-one-family refit, lowest share (drops `mlab`)	46%	25%	49%
Drop-one-family refit, highest share (drops `spn_cryptanalysis`)	47%	31%	49%
Probit link instead of logistic	46%	23%	49%
Logistic with asymptote $\bar{p} = 0.95$ (refit)	46%	30%	49%
Asymptote $\bar{p} = 0.90$	46%	27%	49%
Asymptote $\bar{p} = 0.85$	45%	20%	49%
Task weights and value			
Equal task weights within occupation	46%	27%	49%
Importance × relevance weights	46%	27%	50%
Importance × relevance weights, supplemental tasks admitted (no core filter)	46%	26%	47%
Importance × relevance weights, tasks with importance > 2.5 and relevance > 50%	46%	26%	48%
Frequency-implied weights	45%	28%	51%
Hours-implied weights (duration × annual frequency)	44%	18%	51%
Employment weights, no wages	44%	29%	45%
Excluded set			
Physical filter only: live-interaction activities in scope (previous baseline)	60%	31%	49%
Selling or influencing others kept in scope	47%	27%	49%
Non-physical set = Eloundou et al. exposure label > 0	49%	24%	49%
Excluded only if every work activity is physical or live interaction (lenient)	52%	31%	49%
Mixed tasks split: share $\theta = 0.25$ of value at duration θD	48%	29%	49%
Mixed tasks split: $\theta = 0.5$	49%	30%	49%
Mixed tasks split: $\theta = 0.75$	51%	30%	49%
Also excluded if any work activity is inspecting (task level)	44%	25%	49%
In-scope set = occupations Dingel and Neiman class as teleworkable	38%	16%	49%
Baseline and Dingel–Neiman teleworkable occupation (occupation-level rule)	26%	12%	49%
and not close contact (O*NET physical proximity < 4)	24%	11%	49%
and not close contact (physical proximity < 4.5)	25%	11%	49%
Observed-use definition (expected success unchanged by construction)			
Observed only if share ≥ 0.01%	46%	27%	24%
Observed in Claude.ai only (API traffic left out)	46%	27%	44%
Change of universe (different denominator; not part of the one-change range)			
Observed-duration tasks only (denominator changes)	62%	28%	100%
Observed-duration tasks only, Nov 2025 and Feb 2026 AEI releases	61%	44%	100%

Table 14: Leave-one-out check of the duration imputation. Each observed task is re-imputed from the other observed tasks by the baseline rule, at the level the rule would use for it. $\hat{D}_t/D_t$ is the ratio of imputed to observed duration. Value shares are of the wage bill in observed-duration tasks.

Imputation level	Tasks	Share of observed value	Median $\hat{D}_t/D_t$	Mean log ratio, value-weighted	SD of log ratio	Within factor 2
Detailed work activity	2,736	97%	1.00	+0.02	0.49	87%
Intermediate work activity	69	2%	1.57	+0.94	0.78	57%
Generalized work activity	9	0%	2.61	+1.38	0.84	44%
All observed tasks	2,814	100%	1.00	+0.04	0.50	86%

Table 15: Observed use under alternative definitions. Share of the wage bill in tasks meeting the definition; denominator all tasks. Top block: threshold on the task's share of Claude.ai or API conversations in a window. Bottom block: number of the seven windows in which the task was published at all on either platform.

Definition of "observed"	Tasks	May 2026 window	Any window, Dec 2024 to May 2026
Published at all (baseline)	6,754	30%	49%
Share of conversations $\geq 0.01\%$	2,667	16%	24%
Share $\geq 0.05\%$	873	6%	10%
Share $\geq 0.1\%$	492	3%	6%
Published in at least 1 of the 7 windows	6,754		49%
Published in at least 2 of the 7 windows	5,783		44%
Published in at least 4 of the 7 windows	3,583		26%
Published in at least 7 of the 7 windows	964		11%

G Occupation-Level Exposure and Time Series

Figure 12 plots every 6-digit occupation that can be done remotely and without close contact by its expected-success share of task value under Mythos Preview (early) and the share of its task value that has appeared in Claude usage. The 63 per cent of employment in other occupations has an expected-success share of zero and is not shown. Office, business, sales and computer occupations cluster at the right, with high capability and high use; large occupations with high capability but use well below the economy-wide share, such as general and operations managers, make up much of the "inside the horizon, not yet observed" cell of Figure 5(b). Table 16 gives the employment shares behind Section 5.1, and Table 17 the capability and use measures at each AEI data window.

Figure 13 plots the correlations of Section 5.2, on the baseline filter. The agreement in panels (a) to (c) is largely the excluded set, which both measures largely respect; panels (d) to (f) condition it away by computing both measures over tasks in scope only. Panel (f) supplies the channel: the occupations the rubric scores as most exposed are the ones whose delegated tasks are longest, and the success curve falls with length. Section 5.2 discusses why the two measures diverge. Two cautions: the vertical range in panels (d) and (e) is narrow, so the rank correlation turns on small differences; and the occupations at the top left of panel (e) are mostly occupations with a large excluded share whose small in-scope remainder happens to be short.

Table 16: From tasks to jobs. Share of employment (153.4 million jobs) in occupations whose own coverage measure is at least the threshold. Excluded tasks, and every task of an occupation that cannot be done remotely (Dingel and Neiman, 2020) or needs close physical contact, count against the occupation (probability zero); the in-scope measure drops them from the occupation's denominator. Geometric mean: CES aggregator of task success probabilities with substitution parameter $\rho \to 0$, weighted by task value. Weakest link: the lowest success probability among the occupation's tasks with importance at or above its median. Observed use is not conditioned on.

Occupation-level measure	At least	Claude Opus 4.6	Mythos Preview (early)
Expected-success share of the occupation's task value (additive)	50%	15%	19%
	75%	0%	1%
	90%	0%	0%
Share of the occupation's task value shorter than the 80% horizon	50%	0%	2%
	75%	0%	0%
	90%	0%	0%
Share of the occupation's task value shorter than the 50% horizon	50%	24%	26%
	75%	12%	13%
	90%	4%	8%
Expected-success share of the occupation's *in-scope* task value	50%	37%	37%
	75%	1%	21%
	90%	0%	0%
Geometric mean of task success (CES, $\rho \to 0$)	50%	1%	1%
	75%	0%	0%
	90%	0%	0%
Weakest link: lowest success probability among above-median-importance tasks	50%	3%	3%
	75%	0%	0%
	90%	0%	0%

Table 17: Capability and observed use at each AEI data window. Frontier = longest 50 per cent horizon among models released by the window's last day. Denominator: all tasks. "< p80" = share of the wage bill in tasks in scope shorter than the frontier model's 80 per cent horizon.

AEI window	Frontier model	Released	p50 (h)	p80 (h)	Expected success	< p80	Observed, window	Observed, cumulative
Dec 24–Jan 25	o1	5 Dec 2024	0.6	0.1	14%	0%	27%	27%
Feb–Mar 2025	Claude 3.7 Sonnet	24 Feb 2025	1.0	0.2	17%	0%	25%	30%
Aug 2025	GPT-5	7 Aug 2025	3.4	0.6	30%	3%	25%	34%
Nov 2025	Gemini 3 Pro	18 Nov 2025	3.7	0.9	32%	7%	30%	36%
Feb 2026	Claude Opus 4.6	5 Feb 2026	12.0	1.2	40%	8%	30%	37%
Apr 2026	Mythos Preview (early)	7 Apr 2026	17.4	3.1	46%	27%	27%	47%
May 2026	Mythos Preview (early)	7 Apr 2026	17.4	3.1	46%	27%	30%	49%

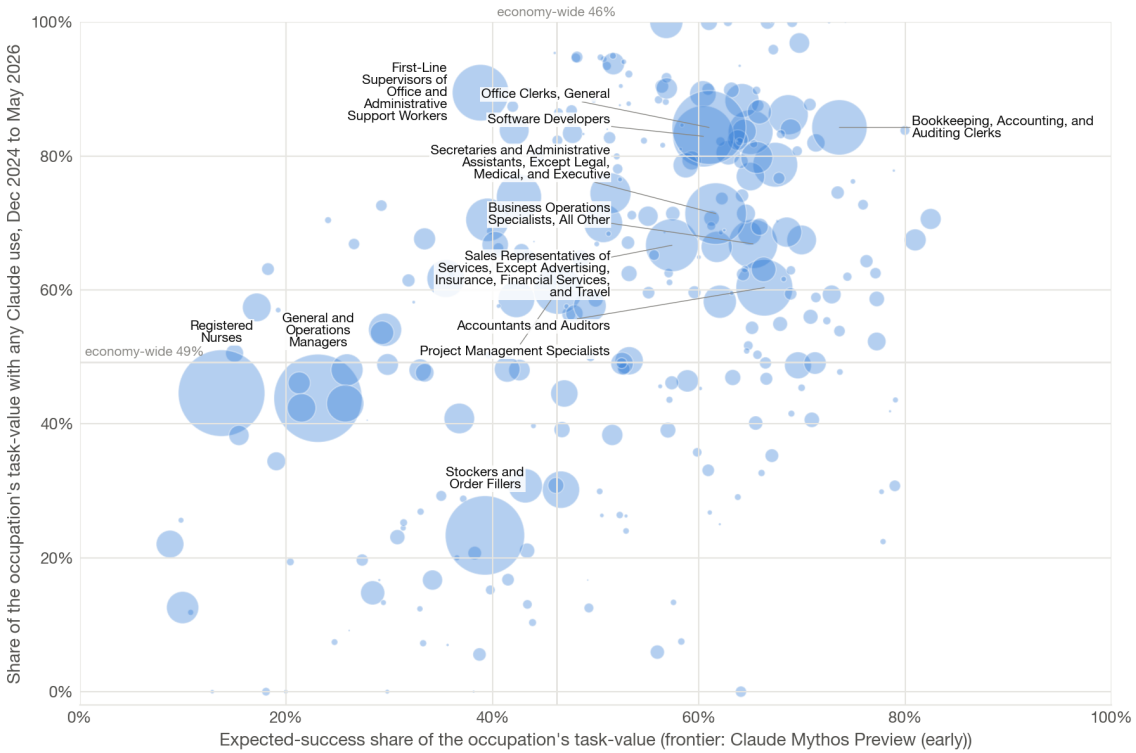


Figure 12: One bubble per 6-digit occupation that can be done remotely and without close contact (285 occupations), area proportional to employment. Horizontal axis: expected-success share of the occupation's task value under Mythos Preview (early), excluded tasks counting as zero. Vertical axis: share of the occupation's task value with a published Claude.ai or API usage share in any AEI release, December 2024 to May 2026. Grey lines: economy-wide values on the baseline filter. The 12 largest occupations shown are labelled.



Figure 13: The comparison with rubric-based exposure measures described in Section 5.2, on the baseline filter. One bubble per 6-digit occupation, area proportional to employment, shade the excluded share of the occupation's task value. Each panel prints the unweighted Spearman rank correlation across the 782 occupations and its employment-weighted Pearson counterpart. Panels (a) to (c) use the additive expected-success share, in which excluded tasks enter at probability zero. Panels (d) to (f) compute both measures over the occupation's tasks in scope only; the expected-success share is then a function of task duration alone. Panel (f) plots exposure against duration. Eloundou exposure is the value-weighted mean of the task-level β labels within the occupation; median duration is wage-weighted; frontier = Mythos Preview (early).

H External Benchmarks and the Horizon Map

The expected-success share depends on the success curve only through its 50 per cent horizon h and slope b, since $P(D) = \sigma(b \log_2(D/h))$ with $b < 0$. Figure 14 maps any (h, b) to the share; every METR model falls on the surface at its own slope. Holding the slope fixed, dividing the horizon by a factor k is equivalent to multiplying every task duration by k, and at METR's doubling time it corresponds to $\log_2 k$ doubling periods of frontier progress.

Each external benchmark sets its own standard of success. For each, I find the factor k at which METR's curve for the same model reproduces the benchmark's observed success rate at the benchmark's task lengths, and apply the same factor to the frontier curves (Table 18). GDPval's gold-set tasks take experts a median of 5 hours, and I treat their durations as log-normal with the published quartiles (Patwardhan et al., 2025). On those lengths METR's curve for Claude Opus 4.1 predicts 30 per cent success, against a win-or-tie rate of 47.6 per cent, so the win-or-tie standard implies a longer horizon. The AEI usage horizons come from Appel et al. (2026), who find that Claude succeeds on half of API tasks of about 3.5 hours and half of Claude.ai tasks of about 19 hours, against METR's 2 hours for the same model. For code judged by repository maintainers rather than by tests, Whitfill et al. (2026) estimate a horizon of about 8 minutes for Claude Sonnet 4.5, against 50 minutes on the automated grader, which they describe as roughly a 7-fold overstatement. The in-context rows apply the 5- to 18-fold gap between METR's contractors and repository maintainers (Kwa et al., 2025) to every duration. METR's cross-domain horizons use METR's own estimates by benchmark (Kwa and Cheng, 2025); several assume a slope of 0.6.

These translations are illustrations, not alternative estimates. Each assumes that the whole gap between two standards acts like a change in task length, with METR's slope held fixed. If part of the gap is a quality standard that short tasks fail as well, a shift in the level of the curve would fit better and would penalise short tasks less. For the AEI usage horizons, the curves I fit to the AEI's own success labels do not identify a 50 per cent horizon on Claude.ai and place it at 1.0 to 1.3 hours on the API, against 3.5 hours in Appel et al. (2026) (Appendix I). Those rows combine a published ratio of horizons with METR's slope, which the labels do not show.

I do not translate the Remote Labor Index (Mazeika et al., 2025). Its five leaderboard entries that match a METR curve complete 1.2 to 4.2 per cent of 240 freelance projects, where METR's curves predict 30 to 51 per cent. But the benchmark reports one rate per model and only the mean (28.9 hours) and median (11.5 hours) project length, and 69 per cent of its projects are in video, audio, design, CAD and game development, a small part of the wage bill. Read as a shift in the horizon, the gap would also imply that the same agents fail most tasks of an hour, where the AEI labels and GDPval both show success rates of about one half or more.

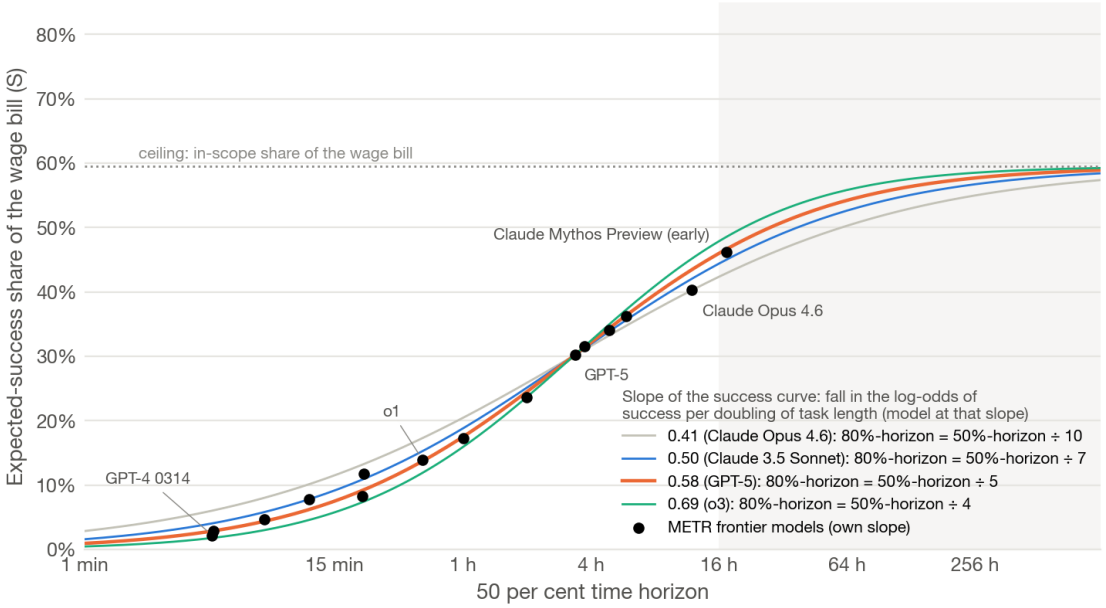


Figure 14: The expected-success share of the wage bill as a function of the 50 per cent time horizon, for four values of the slope of METR's success curve. The slope is the fall in the log-odds of success for each doubling of task length: near the 50 per cent horizon, one doubling lowers the probability of success by about a quarter of the slope, 10 to 17 percentage points across these values. It also sets how far below the 50 per cent horizon the 80 per cent horizon lies: a quarter of it at a slope of 0.69 (o3), a tenth at 0.41 (Claude Opus 4.6). The four values span the frontier models, and each is one frontier model's own slope. Points: METR's frontier models at their own horizon and slope. Grey band: horizons METR describes as unreliable. Dotted line: the ceiling set by the in-scope share of the wage bill.

Table 18: The expected-success share under alternative standards of success and units of human time, as illustrations (see text). Each row divides the frontier model's 50 per cent horizon by the factor shown, holding its slope; months = $\log_2$ of the factor times the doubling time of our fit (4.2 months). †: derived from published figures rather than from data (Appendix A).

Standard of success or unit of human time	Horizon divided by	Months of progress	Expected success	
			Opus 4.6	Mythos
AEI usage, Claude.ai†	0.11	−14	53%	57%
GDPval, wins or ties against experts†	0.38	−6	47%	52%
AEI usage, API†	0.57	−3	44%	50%
Baseline: METR's curves as estimated	1	0	40%	46%
All durations doubled	2.0	+4	35%	40%
SWE-bench Verified (METR cross-domain)	3.1	+7	31%	36%
In-context workers, 5× faster (METR)†	5.0	+10	27%	31%
Maintainer review, not tests (METR)†	7.0	+12	25%	27%
In-context workers, 18× faster (METR)†	18	+18	17%	18%
WebArena (METR cross-domain)	21	+18	17%	16%
Computer use, 40× shorter (METR)†	40	+23	12%	11%
OSWorld (METR cross-domain)	66	+26	10%	8%
Computer use, 100× shorter (METR)†	100	+28	8%	6%

I Task Success and Task Length Outside METR's Suite

The January and March 2026 AEI releases report, for each O*NET task and platform, the number of conversations that Claude classified as successful or unsuccessful, alongside the mean human time of those conversations. I keep tasks with at least ten classified conversations, which covers about 0.8 million conversations per window and platform, and regress success on $\log_2$ of human minutes in a conversation-weighted binomial model, pooled and with occupation or intermediate work activity fixed effects (Table 19). For tasks observed in both releases, a model with task and release fixed effects asks whether the same task succeeds less often in the release in which its delegated piece is longer: on the API it does, at 0.37 per doubling over 1,411 tasks, while on Claude.ai the slope is −0.08. Figure 15(a) plots success by duration bin on the API in the February 2026 window against the METR curve of Claude Opus 4.5. For tasks under 15 minutes the curves predict 89 to 96 per cent success against 58 to 74 per cent observed; for tasks of four to sixteen hours they predict less success than observed on Claude.ai. Across occupations, the rank correlation between the expected-success share of value in scope and observed success is 0.02 to 0.24.

Using the pooled AEI curves in place of METR's would not raise the headline share. On the baseline task frame, the API curves give an expected-success share of 25 to 26 per cent and the nearly flat Claude.ai curves 40 to 42 per cent, against 46 per cent under Mythos Preview (early). A flatter gradient raises success on long tasks but lowers it on short ones, and these curves describe the models that served the traffic, not the latest frontier. Against Claude Opus 4.6, the latest of those models (40 per cent), the API curves give less and the Claude.ai curves slightly more. Neither is an alternative estimate, because the labels measure success given delegation rather than autonomous completion.

GDPval reports success by task length on a fixed set of tasks, which users do not select (Patwardhan et al., 2025). Its gold-set tasks were written by professionals, and each deliverable is graded blind against an expert's. The paper reports win rates for four bins of expert time; I place each bin at its median under a log-normal fitted to the published quartiles and fit a logistic curve to the four rates. The fall in the log-odds per doubling is 0.20 to 0.24 across Claude Opus 4.1, GPT-5 and o3, against 0.58 to 0.69 on METR's suite for the same models (Figure 15(b)). The tasks are one-shot and well specified, the times are self-reported, and four bins give a coarse gradient. The win rate in this breakdown probably counts only deliverables rated better than the expert's, whereas the calibration in Appendix H uses wins and ties, so the gradient and the level rest on slightly different standards.

Figure 15(c) shows the curves that Mertens et al. (2026) fit to workers' evaluations of single responses, for two Claude models that METR also measured. Their curves are much flatter than METR's and move up over time rather than out (Section 5.4).

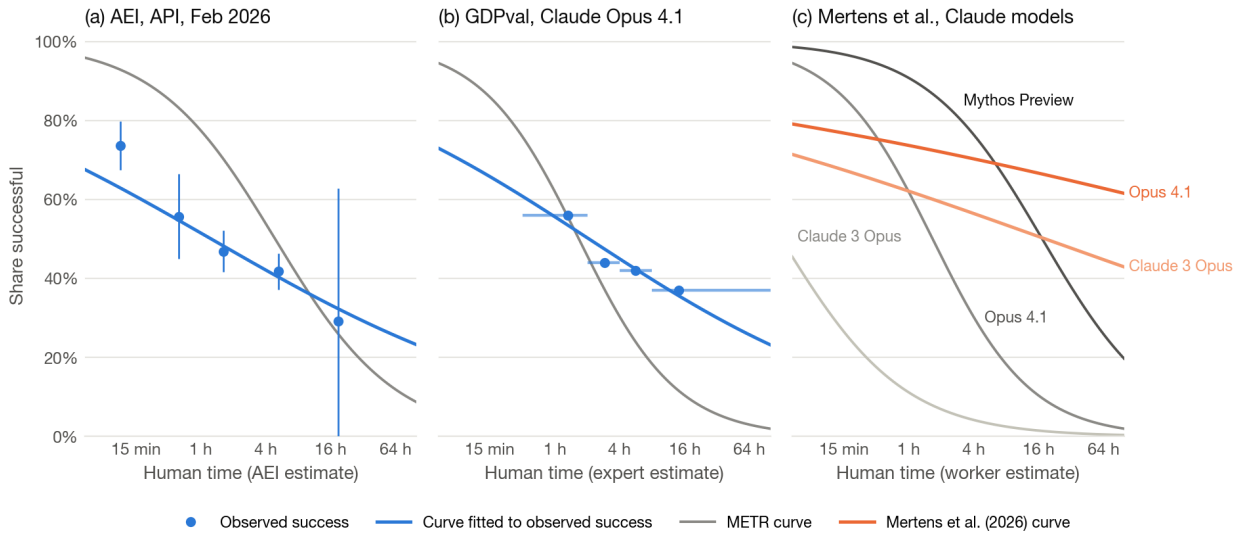


Figure 15: Success against human time outside METR's suite, against METR's curves for the same Claude models (grey). Blue: observed success and a logistic curve fitted to it. (a) Share of API conversations in the AEI's February 2026 window classified as successful, by the task's mean human time, for tasks with at least ten classified conversations; bars are 95 per cent intervals clustered by task. Grey: Claude Opus 4.5, the latest Claude model with a METR curve released before the window (Claude Opus 4.6 was released on its first day). (b) GDPval win rates against an expert for Claude Opus 4.1, by bin of the expert's time (bars span each bin; points at the bin medians) (Patwardhan et al., 2025). Grey: Claude Opus 4.1. (c) Orange: the curves that Mertens et al. (2026) fit for Claude 3 Opus (March 2024) and Claude Opus 4.1 (August 2025), the share of each model's responses that workers judge usable without edits, by the worker's estimate of the task's length (their Figure A.5, human time in minutes; Appendix A). Grey: METR's curves for the same two models and for Mythos Preview (early), released in April 2026, which they did not evaluate (their latest model was released in August 2025). Darker shades are later models. Their task-level points are not public, so only curves are shown. Fall in the log-odds of success per doubling of human time: (a) fitted 0.19, METR 0.54; (b) fitted 0.21, METR 0.66; (c) Mertens et al. (2026) 0.12 and 0.08, METR 0.53, 0.66 and 0.56.

Table 19: AEI task success and human time. Binomial regressions of classified success on $\log_2$ human minutes, weighted by conversations; heteroskedasticity-robust standard errors in parentheses. The slope is reported as the fall in the log-odds of success per doubling, the same units as METR's slopes (0.41 to 0.66 for the Claude models serving these windows). Within occupation: 6-digit SOC fixed effects. Within activity: fixed effects for the intermediate work activity of the task's first detailed work activity. Within task: tasks observed in both releases on the same platform (2,225 on Claude.ai, 1,411 on the API), with task and release fixed effects.

			Mean	Fall in log-odds of success per doubling			
Platform	Window	Tasks	success	Pooled	Within occupation	Within activity	Within task
Claude.ai	Nov 2025	2,388	68%	0.04 (0.05)	0.10 (0.03)	0.05 (0.04)	−0.08 (0.07)
Claude.ai	Feb 2026	2,551	71%	−0.04 (0.07)	0.02 (0.03)	−0.06 (0.04)	
API	Nov 2025	1,666	49%	0.19 (0.07)	0.12 (0.05)	0.17 (0.05)	0.37 (0.06)
API	Feb 2026	1,680	50%	0.19 (0.09)	0.13 (0.05)	0.21 (0.04)	

J In-Person Rules

The Dingel–Neiman rule classifies an occupation as teleworkable unless any of fifteen conditions holds, such as rarely using email, working outdoors every day, or rating physical activities, handling objects, dealing with the public or inspecting equipment as very important (Dingel and Neiman, 2020). I transcribe the conditions from their published code and apply them to the Work Context and Work Activities files of O*NET 30.2. On the 720 occupation codes the two O*NET releases share, the rebuilt classification agrees with theirs on 94.6 per cent; the rest reflects revised O*NET ratings. On this paper's weights, teleworkable occupations hold 51 per cent of the wage bill and 39 per cent of employment, against the 46 and 37 per cent that Dingel and Neiman report for 2018. Tasks in occupations without Work Context data (3.7 per cent of the wage bill) are kept. Table 20 gives the top ten occupations on the baseline filter, which the Dingel–Neiman rule removes from the top of the ranking. For the job-level results I add one condition that their code computes but leaves out of the rule: an O*NET physical-proximity rating (Work Context 4.C.2.a.3) below 4. It excludes school teachers, teaching assistants, childcare workers, food-service supervisors, ushers and gambling dealers, among others; a few exclusions are debatable, such as tutors and air traffic controllers. A threshold of 4.5 gives similar results (Table 13).

Table 20: As Table 5, on the baseline task rule (only excluded tasks count as zero).

SOC	Occupation	Employment (thousands)	Expected success	Weakest link	Observed use	Median duration (h)
45-4023	Log Graders and Scalers	3	85%	79%	0%	1.4
43-4021	Correspondence Clerks	4	85%	74%	85%	1.7
45-2011	Agricultural Inspectors	14	84%	83%	51%	2.4
29-2072	Medical Records Specialists	195	82%	0%	71%	1.5
41-2012	Gambling Change Persons and Booth Cashiers	22	82%	76%	17%	3.4
43-9041	Insurance Claims and Policy Processing Clerks	214	81%	69%	67%	3.4
43-4131	Loan Interviewers and Clerks	165	80%	72%	57%	4.3
31-9094	Medical Transcriptionists	42	80%	70%	84%	3.9
13-1032	Insurance Appraisers, Auto Damage	12	79%	68%	45%	3.0
17-1021	Cartographers and Photogrammetrists	14	79%	71%	44%	2.8
31-1120	Home Health and Personal Care Aides	4,306	23%	0%	50%	1.5
41-2031	Retail Salespersons	3,898	48%	0%	73%	2.1
35-3023	Fast Food and Counter Workers	3,854	24%	0%	22%	0.6
11-1021	General and Operations Managers	3,503	23%	0%	44%	7.0
29-1141	Registered Nurses	3,380	55%	0%	45%	2.9

K Checking Costs

Suppose that delegating a task to the agent costs the worker checking time vD_t, that a detected failure means the worker redoes the task, and that an undetected or costly failure adds a further cD_t with probability $1 - P_t$. The expected time saved on a delegated task is then $D_t[P_t - v - c(1 - P_t)]$, and a worker delegates only when this is positive. Table 21 reports the resulting net share of the wage bill saved, for which $v = c = 0$ gives the expected-success share. Anthropic's reliability adjustment of its own productivity estimate, from 1.8 to 1.0–1.2 percentage points a year (Appel

et al., 2026), corresponds to v of 0.23 to 0.35 and a net share of 22 to 31 per cent. Only 8.0 per cent of the wage bill is in tasks with $P \geq 0.9$ under Mythos Preview (early), and 1.4 per cent with $P \geq 0.95$.

Table 21: Net share of the wage bill saved after checking costs. v: checking time as a share of the task's human time. c: additional cost of a failure as a share of the task's human time ($c = 0$: failures are caught and redone; $c = 1$: each failure costs one further task). Tasks in scope only; excluded tasks count as zero.

	Mythos Preview (early)		Claude Opus 4.6	
Checking time v (share of the task's human time)	$c = 0$	$c = 1$	$c = 0$	$c = 1$
0.00	46%	33%	40%	21%
0.10	40%	27%	34%	16%
0.25	31%	18%	25%	9%
0.50	16%	7%	11%	2%